The Subsector Composition of Portugal's General-Government Balance, 1977–2025

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Abstract

A general-government balance is reported as a single number, although it is the sum of institutionally distinct subsectors with different revenue bases, expenditure mandates and borrowing constraints. We assemble a harmonised annual panel of net lending and net borrowing for Portuguese General Government and its three subsectors over 49 years, 1977–2025, and decompose the aggregate exactly into its subsector components and their annual changes.

The composition is asymmetric and stable. The aggregate balance is positive in 4 years (2019, 2023, 2024 and 2025); the Central Government balance is negative in all 49; the Social Security balance is positive in 43, with a longest run of 24 years. General Government and Central Government track each other closely within each statistical regime, correlating at 0.973 before the 1995 splice and 0.976 after it. The empirically informative feature of the 4 surplus years is that the combined non-Social-Security balance is negative in all of them; that the Social Security balance then exceeds it in absolute size follows from the identity.

Two findings cut against readings the aggregate invites. First, a persistently negative Central Government B.9 does not imply a persistently negative primary balance: excluding interest, the primary balance is positive in 15 of the 45 years with detailed accounts — 9 of 19 before the 1995 splice and 6 of 26 after it. Second, the ESA 2010 Social Security Funds sector and the budget-execution systems published by the Portuguese Public Finance Council are different accounting objects whose balances differ in every one of the 7 years they overlap, so quoting one for the other misstates the figure that enters the national accounts.

Relating the Social Security accounts to a quantity outside them, the 2025 rise of 2,468 million euro in social contributions decomposes exactly into 1,870 million from the aggregate employee wage bill, 560 million from the contributions-to-wage-bill ratio and 38 million from their interaction. The wage bill is a reference base rather than the integral base of the levy, and the ratio is not a statutory rate.

Applying the same definitions to 28 European reporters shows the composition is distinctive in a specific respect. A permanently deficit-running central tier is common — 9 reporters record one in every year — but Portugal's Social Security surplus is in the upper tail on both frequency and size, and it is the only reporter whose every aggregate-surplus year combines that surplus with a negative non-Social-Security balance, a composition found in 27% of European surplus years.

Magnitudes are regime-specific: the aggregate averages -6.43% of GDP before the 1995 statistical splice and -4.04% after, so no level statistic is pooled across it. The analysis is descriptive and accounting-based; it decomposes recorded quantities and identifies no causal effect.

Keywords: general government; institutional subsectors; social security funds; accounting decomposition; fiscal statistics; Portugal
JEL: H61, H62, H55, E62

Contents

1 Introduction

A country's fiscal position is normally reported as one number. Portugal recorded a general-government balance of 2,059 million euro in 2025, or 0.67% of GDP, and it is that figure which enters fiscal rules, credit assessments and public debate.

The number is an aggregate of institutionally distinct parts. Under ESA 2010 the general government sector S.13 is the consolidation of central government (S.1311), state government (S.1312), local government (S.1313) and social security funds (S.1314) (Eurostat, 2013). These are not administrative subdivisions of one budget. They have different revenue bases — general taxation in one case, earmarked social contributions in another — different expenditure mandates, and different legal constraints on borrowing. Their balances are therefore determined by different things, and there is no reason to expect them to move together.

Aggregation obscures that. In 2025 Portugal's positive aggregate balance was the sum of a Central Government balance of -5,636 million euro, a Regional and Local balance of 630 million, and a Social Security Funds balance of 7,065 million. The headline figure and the composition behind it support quite different descriptions of the same year. The aggregate is what fiscal targets attach to and what receives most of the attention; the subsector decomposition, although published and regularly discussed by fiscal institutions, has attracted much less attention as a long-run object.

1.1 Research questions

This paper addresses two questions.

1. **How has Portugal's general-government balance been composed across institutional subsectors since 1977, and how persistent are the contributions of Central Government, Regional and Local Government, and Social Security Funds?**

2. **How is the Social Security Funds balance composed, which account movements produce its annual changes, and how does its national-accounts boundary relate to the internal Social Security budget systems?**

The second question is posed at the level of accounts rather than of causes, and it is answered in two steps. We first decompose the change in the balance into the revenue and expenditure movements that produced it, which locates the change inside the accounts. We then take one step outside them, relating contributions to the aggregate employee wage bill that is their principal observable reference base (Section 7). That second step names a quantity the contributions moved with, and quantifies how much of their movement it accounts for. It does not identify what caused either to move, which would require a model of entitlement rules, labour supply and demographics that this paper does not build.

Both are descriptive questions about recorded quantities, and we answer them as such. The central object is an accounting identity,

$$B_t^{GG} = B_t^{C} + B_t^{RL} + B_t^{SSF},$$

which holds by construction. Decomposing it reallocates an observed number; it estimates nothing and identifies nothing. We are explicit about this throughout, because the decomposition invites a counterfactual reading — that the aggregate balance would have been different under a different institutional arrangement — which the identity cannot support.

1.2 Related work and where this paper sits

Three strands of existing work bear on these questions, and each leaves a different part of them open.

The first is the statistical framework itself. ESA 2010 defines the subsectors and requires member states to report deficit and debt for each of them, and Eurostat and the ECB publish the resulting series (Eurostat, 2013, 2024b, 2025; European Central Bank, 2025). The composition of the aggregate balance is therefore not hidden: it is published. What the framework provides is a definition and a data supply, not an analysis of how a particular country's composition has behaved over time.

The second is the literature on Portuguese fiscal sustainability, which works predominantly at the aggregate level and over long horizons. Marinheiro (2006) tests the sustainability of Portuguese fiscal policy on more than a century of annual data, using stationarity and cointegration tests on aggregate revenue, expenditure and debt. That question — whether the aggregate path satisfies a solvency condition — is a different question from ours, and it is answered without decomposing the aggregate.

The third concerns Portuguese Social Security financing specifically. Cardoso (2019) examines the equity and sustainability of the system's financing, noting that social-protection entitlements follow a dynamic that at times diverges from the cyclical path of the revenue that funds them; the Portuguese Public Finance Council has also examined the Social Security Financial Stabilisation Fund in the same context (Portuguese Public Finance Council, 2024). This literature is concerned with the system's own finances and its long-run adequacy, rather than with the arithmetic by which that system's balance enters the national aggregate.

The Council's monitoring publications are more directly relevant still, and they bear on what this paper may claim as new. Reporting on 2025, Portuguese Public Finance Council (2026b) records a general-government surplus alongside a Central Government deficit that widened for a second consecutive year, and states that the Social Security Funds continued to be a key factor in the aggregate result; the companion report on the Social Security accounts attributes the improvement in the system's budget balance to revenue growth exceeding expenditure growth, with social contributions the main driver (Portuguese Public Finance Council, 2026c). **The recent pattern this paper documents is therefore already identified, for the recent years, by Portugal's independent fiscal institution.** We are explicit about that rather than presenting it as a discovery. It also provides an external check on our extraction: the Council's figure for the 2025 Social Security budget balance is 6,657 million euro, which is exactly the total we obtain by summing the three budget systems.

A fourth strand is relevant less as a precedent than as a caution. Work on stock-flow adjustments has shown that the reported balance is an incomplete account of what happens to debt: von Hagen and Wolff (2006) find that debt accumulates by more than accumulated deficits and read the residual as evidence of accounting responses to fiscal rules, while Seiferling (2013) argues on integrated public-finance data that the residuals are smaller than previously assumed and are not well explained by fiscal transparency. We take no position in that debate. We take from it the discipline that a balance is a recorded accounting quantity whose relation to anything else has to be demonstrated rather than assumed — which is why this paper stops at the accounting and says so.

Taken together, these strands locate what is left open. The framework supplies the subsector definitions and the data. The academic literature on Portugal works at the aggregate level over long horizons. The Council documents the subsector composition in detail, but for the recent years and as monitoring rather than as a long-run series. What we have not found is a systematic treatment of the composition itself over a window of this length: the persistence of each subsector's sign, the regime-dependence of the magnitudes, the attribution of annual movements, and the relation between the two Social Security accounting boundaries.

We state that as a gap in the work we have found, not as a claim of priority. The series are published; a competent user of the Eurostat or Council data could construct parts of what follows; and the recent pattern is, as noted, already public. The contribution lies in the assembly over five decades, the harmonisation across a methodological break, the discipline applied to the resulting statistics, and the reproducibility of the whole.

1.3 Contribution

We make four contributions.

1. **A harmonised long-run panel, and a validation layer that says what it is worth.** An annual panel of net lending and net borrowing for General Government and its three subsectors covering 1977–2025, built by splicing the Banco de Portugal / INE historical long series to the modern national accounts. The **1995** methodological boundary is retained as a documented splice, both vintages of the overlapping year are kept, and nothing is chained, smoothed or calibrated across it. Alongside it we separate two checks that are routinely conflated: identity closure asks whether an extraction is arithmetically self-consistent, source agreement asks whether two independently published sources report the same number, and the first cannot establish the second. Here the identities close to 1.00 million euro while two published sources still disagree by 66.8 million on one subsector-year.

2. **An exact decomposition of the balance and of its movements, at three levels.** The aggregate balance and each of its annual changes decompose into subsector components that close to the sources' rounding, with changes additionally scaled by current-year GDP so that episodes forty years apart are comparable without turning the decomposition into a ratio that no longer adds up. The largest episodes are then attributed hierarchically: to a subsector, to that same subsector's revenue and expenditure, and to the components of those accounts. Persistence is reported alongside, with magnitudes computed inside each statistical regime rather than pooled across the splice — not a presentational choice, since the aggregate balance averages -6.43% of GDP before 1995 and -4.04% after, and the pooled -4.91% describes neither period.

3. **Social Security: two accounting boundaries, and a contribution base.** The ESA 2010 Social Security Funds sector and the budget-execution systems published by the CFP are different accounting objects; we report them side by side, never add them, and measure the gap, which is non-zero in every one of the 7 years they overlap. Within the national-accounts sector, each annual balance change splits into the account movements

that produced it, and those contributions are then related to the aggregate employee wage bill that is their principal observable reference base — a reference base and not the integral base of the levy, since national-accounts contributions include imputed amounts and bases the wage bill does not contain. That last step is what takes the Social Security findings from an accounting location to a named quantity outside the fiscal accounts.

4. **A European benchmark.** The same definitions applied to 28 reporters over 1995–2025, which is what turns "this is how Portugal behaves" into a statement about whether that behaviour is unusual. A single-country study cannot make that distinction, and the answer turns out to differ across the findings.

1.4 Principal findings

The composition is asymmetric and it is stable. Over 49 years the aggregate balance is positive in 4 (2019, 2023, 2024 and 2025). The Central Government balance is negative in all 49 observations. The Social Security balance is positive in 43, with a longest unbroken run of 24 years. Regional and Local Government is the only subsector that changes sign with any regularity, and it is small in both directions. The aggregate tracks Central Government closely in both statistical regimes: their GDP ratios correlate at 0.973 before the 1995 splice and 0.976 after it, with median absolute gaps of 0.53 and 0.55 percentage points of GDP. The correlation is computed inside each regime and never through the splice, where part of the co-movement would be a vintage change.

In each of the 4 years with a positive aggregate balance, the combined non-Social-Security balance is negative. That is the empirical content of the observation, and it is worth isolating: once $B_t^{GG} > 0$ and $B_t^{nonSSF} < 0$ are both established, it follows from the identity that $B_t^{SSF} > |B_t^{nonSSF}|$, so the third condition is arithmetic rather than a separate finding. What remains to be quantified is the magnitude. In 2025 the non-Social-Security balance was -5,006 million euro against a Social Security balance of 7,065 million, an offset ratio of 1.411.

Magnitudes are regime-dependent and must be read that way. The Central Government balance averages -6.61% of GDP over 1977–1994 and -4.67% over 1995–2025; the Social Security balance rises from 0.46% to 0.72%.

A persistently negative headline balance does not imply a persistently negative primary balance. Central Government records a negative B.9 in every year of the panel, but its primary balance — the balance excluding interest — is positive in 15 of the 45 years for which the detailed accounts exist, from 1986 to 2025: 9 of 19 years in the historical regime and 6 of 26 in the modern one. Interest is the arithmetic separating the two series, and it is much the larger burden in the earlier period, averaging 5.52% of GDP historically against 3.19% in the modern regime. We use "primary balance" throughout and avoid calling it an underlying balance, which would suggest a structural or cyclically adjusted measure that this paper does not compute.

Against the European comparison set, the answer differs across these findings, which is the reason for making it. A persistently deficit-running central tier is *not* distinctive: 9 of 28 reporters record a Central Government deficit in every year they cover. The Social Security surplus *is*, in both frequency (93.5% of years against a median of 68.9%) and size (0.71% of GDP against 0.17%). And the composition of the surplus years is the sharpest difference: across the 22 reporters with any aggregate surplus, only 27% of surplus country-years combine that surplus with a negative non-Social-Security balance, and Portugal is the only reporter for which every surplus year does — on 4 observations, which is few enough that we state the count alongside the claim.

Finally, on the Social Security side, the annual change decomposes into its accounts. In 2025 the balance rose by 1,034 million euro, with social contributions contributing 2,468 million and other revenue 1,087 million, against -2,003 million from social transfers and -518 million from other expenditure. Within the internal budget systems, the movement over the published window is concentrated in the Previdential system, which goes from 2,333 million euro in 2019 to 6,712 million in 2025 while the Citizenship balance stays small in both directions at -55 million.

1.5 Roadmap

Section 2 defines the institutional and accounting objects, and in particular why Central Government is not a synonym for the State in the budgetary sense. Section 3 describes the sources, the harmonisation, the 1995 splice, the vintages and the cross-source validation. Section 4 states the decompositions and the persistence measures. Section 5 presents the results. Section 6 sets out the accounting boundaries and the robustness of the weaker statistics. Section 7 relates Social Security contributions to the aggregate employee wage bill that serves as their reference base. Section 8 places Portugal in the European distribution. Section 9 concludes with what the evidence does and does not establish. Appendix A reports the change-point sensitivity in full and Appendix B the reproduction path.

2 Institutional and Accounting Framework

2.1 The subsectors

Under ESA 2010 the general government sector S.13 consolidates four subsectors (Eurostat, 2013):

- **S.1311, central government**, comprising all government units with a national sphere of competence *with the exception of social security units* (Eurostat, 2024a);
- **S.1312, state government**, which does not apply to Portugal;
- **S.1313, local government**, which in the Portuguese case is reported together with the autonomous regions and appears here as Regional and Local Government;
- **S.1314, social security funds**, comprising all social security units *regardless of the level of government that operates or manages the scheme* (Eurostat, 2024b).

The two emphasised clauses matter for everything that follows, and they are the source of a recurring misreading.

Central Government in the national accounts is not the State in the budgetary sense. The statistical subsector excludes social security units by definition, and it includes many entities — public institutes, funds, reclassified public corporations — that no reader of a State Budget would think of as the State. A statement about S.1311 is therefore not a statement about the *Orçamento do Estado*, and the two should not be substituted for one another. Where this paper says Central Government it means S.1311 throughout.

The social security subsector is defined by function, not by tier. S.1314 collects social security units whichever level of government runs them. It is consequently not a level of administration sitting beneath Central Government; it is a functionally delimited sector alongside it.

2.2 The identity

The balancing item is B.9, net lending (+) or net borrowing (–), and it is the difference between total revenue and total expenditure for the sector concerned (Eurostat, 2025). Consolidation across subsectors means the aggregate is their sum:

$$B_t^{GG} = B_t^{C} + B_t^{RL} + B_t^{SSF}. \tag{1}$$

Equation (1) is definitional. In published data it closes only to the rounding of the published series, and we carry that residual as a column rather than assume it away; over the whole panel its largest absolute value is 1.00 million euro, which is the rounding unit of the sources.

Because the subsectors are consolidated, transfers between them net out of the aggregate but not out of the components. A transfer from Central Government to the Social Security Funds worsens B^C and improves B^{SSF} while leaving B^{GG} unchanged. This is central to interpreting the composition: the *location* of a balance within general government depends on the transfer convention, even though the aggregate does not. We return to this in Section 6 and deliberately do not construct a transfer-adjusted balance and call it underlying.

2.3 Two things the identity cannot do

It cannot support a counterfactual. Writing $B_t^{nonSSF} = B_t^C + B_t^{RL}$ and observing that $B_t^{SSF} > |B_t^{nonSSF}|$ in a given year says that two recorded numbers stand in a particular ratio. It does not say what the aggregate balance would have been had the Social Security Funds been organised differently, because the counterfactual would change contribution rates, entitlements and transfers simultaneously, and the identity contains no model of any of them.

It cannot support an attribution of responsibility. A subsector that accounts arithmetically for most of an annual improvement has not been shown to have produced it. The decomposition is silent on intent, on policy and on causation, and we use the word contribution in its arithmetic sense only.

3 Data and Harmonisation

3.1 Sources

The canonical balance panel takes 1977–1994 from the Banco de Portugal / INE historical long series (Banco de Portugal, 2023) and 1995–2025 from INE national accounts distributed through PORDATA (PORDATA, 2026). Detailed revenue, expenditure, interest, investment, Maastricht debt and stock-flow data for the modern period come from the ESA 2010 annual workbooks published by the Portuguese Public Finance Council (Portuguese Public Finance Council, 2026a), which also serve as an independent cross-check on the balance series. Table 1 records each file, the coverage taken from it, its vintage and a digest of the retained copy.

Every workbook is parsed programmatically from the bundled file. No value in this paper was transcribed by hand, including the values in this sentence: they are macros written by the pipeline from the persisted artefacts.

Table 1: Bundled sources, the coverage taken from each, the vintage of the retained file and the first twelve hexadecimal digits of its SHA-256 digest.

Source	Institution	Coverage used	Vintage	SHA-256 prefix
Banco Portugal Long Series	Banco de Portugal / INE	1977-1995 for the historical segment used here	2023-12	d3570ea90f78
PORDATA Balance By Level	INE via PORDATA	1995-2025	2026-04	2a5d54649041
CFP Annual General Government	Portuguese Public Finance Council (CFP)	1995-2025	2026-04-15	8f6b5a2916a4
CFP Annual Subsectors	Portuguese Public Finance Council (CFP)	2000-2025	2026-04-15	4d9e1b07cd38
Eurostat Subsector Balances	Eurostat	1995-2025, EU and EFTA reporters, B9 by subsector	2026-07-21	cf4c9408c237
INE Wage Bill	Statistics Portugal (INE) via Eurostat	1995-2025, total economy	2026-08-17	e925c41100d8
INE Employment	Statistics Portugal (INE) via Eurostat	1995-2025, total economy	2026-08-17	7595dfe29633
CFP Social Security Detail	Portuguese Public Finance Council (CFP)	2019-2025 systems; detailed 2024-2025 tables	2026-04	d9f28e424ab6

3.2 Vintage

A fiscal series has a vintage as well as a coverage, and the two are different things. The modern national-accounts and CFP files used here are the April 2026 releases. Within them, 2024 and 2025 are flagged **provisional** by the publisher, and that flag is carried through the harmonisation into the canonical panel rather than dropped.

The distinction is not pedantry in this application. The provisional years are the years a reader is most likely to be interested in, and they are the years most likely to be restated. Any statement below about 2024 and 2025 should be read as a statement about the April 2026 vintage of those years.

3.3 Two statistical regimes, not one series

The splice at **1995** separates two source families built on different methodologies. We retain it explicitly. Nothing is chained, smoothed, interpolated or calibrated across it, and two consequences are enforced throughout the paper:

1. no statistic whose value depends on the *level* of the balance — a mean, a segment mean, a regression intercept — is computed across the boundary;
2. no annual change is computed through it, so the 1994-to-1995 change, which would mix a vintage revision with an economic movement, appears nowhere in the results.

1995 exists in both source families, so rather than choosing silently we keep both vintages and report the revision between them. The revisions are small in level terms but non-zero for every subsector, which is the direct evidence that the two regimes are not interchangeable at the level of a single year.

Signs are more robust to the splice than magnitudes — checked, not assumed

A natural claim at this point is that a balance's sign is insensitive to the vintage while its level is not. Stated generally that is false: a methodological revision can move a balance close to zero across it, turning a small surplus into a small deficit.

What can be checked is whether it happens here, and in the overlaps this panel retains it does not. Across the 4 balances observed in both 1995 vintages, the sign is the same in 4. Across the 109 subsector-years in which the PORDATA bridge and the CFP workbook both report a figure, the sign agrees in 109. We therefore treat sign classifications as empirically invariant across the source overlaps available here, and as less exposed to the splice than magnitudes — which is a claim about this panel rather than a general property of fiscal statistics.

3.4 An explicit gap

Detailed revenue and expenditure components are available for General Government continuously from 1977. For the three subsectors they exist for 1977–1995 and 2000–2025 only. The 4 intervening years are absent from both source families and are therefore absent here: they are not imputed, no statistic is computed across them, and every figure drawn from the account panel breaks its line rather than joining 1995 to 2000.

The gap affects the detailed components only. The canonical B.9 panel has an observation for every one of the 49 years and for every subsector, which is why the composition results in Section 5 cover the full window while the revenue–expenditure results do not. The detailed account panel holds 184 sector-year observations against the 4×49 a complete panel would have, and Table 2 states the two coverages separately: reporting a single observation count for a panel that has two would be misleading.

Table 2: Coverage of the detailed account panel. The canonical balance panel is complete; only the detailed components carry the 1996–1999 gap.

Subsector	First year	Last year	Observations
General Government	1977	2025	49
Central Government	1977	2025	45
Regional and Local	1977	2025	45
Social Security Funds	1977	2025	45

3.5 Identity closure is not source agreement

Two different validation questions are easy to conflate, and conflating them overstates what the checks establish.

An **identity** check asks whether an extraction is arithmetically self-consistent: whether B.9 equals the sum of its subsectors, whether revenue minus expenditure reproduces the recorded balance, whether the change in debt reconciles with the balance and the stock-flow adjustment. Such a check can close to numerical precision while both sources are wrong in the same way, because it never consults a second source.

A **source-agreement** check asks whether two independently published sources report the same number for the same year. Identity closure cannot establish it.

Table 3 reports both in one unit. The identities close to 1.00 million euro, the rounding of the published series. The largest disagreement between the PORDATA bridge and the CFP workbook is 66.8 million euro, on Central Government in 2002. That gap is nearly two orders of magnitude larger than any identity residual, which is the concrete reason for reporting the two separately: a paper that reported only the identity residuals would imply a precision the underlying statistics do not have.

Table 3: Identity closure and source agreement, in one unit. The two answer different questions and neither substitutes for the other.

Check	Quantity compared	N	Largest absolute difference (M EUR)	Where
Accounting identity	Aggregate balance minus the sum of its three subsectors	49	1.000	2004
Accounting identity	Revenue minus expenditure minus the recorded balance	184	0.000	Central Government, 1994
Accounting identity	Debt change against minus the balance plus the stock-flow adjustment	108	0.000	2000
Source agreement	PORDATA bridge against the CFP workbook, General Government	31	0.498	1995
Source agreement	PORDATA bridge against the CFP workbook, Central Government	26	66.811	2002
Source agreement	PORDATA bridge against the CFP workbook, Regional and Local	26	0.492	2002
Source agreement	PORDATA bridge against the CFP workbook, Social Security Funds	26	0.470	2019
Vintage revision	1995 modern vintage against the historical vintage, four balances	4	72.465	1995

4 Empirical Framework

Every quantity in this section is an accounting transformation of published data. Nothing is estimated except the piecewise means of Section 6, which are reported with their sensitivity.

4.1 Level decomposition

The identity of Equation (1) is applied year by year. Balances are expressed both in million euro and as a share of contemporaneous nominal GDP, the latter for comparability across five decades of nominal growth.

4.2 Dynamic attribution

Differencing the identity attributes each annual change to the subsectors:

$$\Delta B_t^{GG} = \Delta B_t^{C} + \Delta B_t^{RL} + \Delta B_t^{SSF}. \tag{2}$$

This separates the *level* of a subsector balance from its *contribution to a movement*, which are distinct and which the level series alone cannot distinguish: a subsector can be in persistent surplus and still account for a deterioration.

Ranking episodes on nominal euro would rank them by how recent they are, since nominal GDP in 2025 is orders of magnitude above 1977. We therefore also report each term scaled by current-year GDP:

$$\frac{\Delta B_t^{GG}}{GDP_t} = \frac{\Delta B_t^{C}}{GDP_t} + \frac{\Delta B_t^{RL}}{GDP_t} + \frac{\Delta B_t^{SSF}}{GDP_t}. \tag{3}$$

Because the four terms share one denominator, Equation (3) decomposes exactly, and its residual is Equation (2)'s residual rescaled and nothing more. It is emphatically **not** the change in the balance ratio, $\Delta(B_t^{GG}/GDP_t)$, which also moves with the denominator and does not decompose additively. We use the scaled form for ranking and comparison and the level form for magnitudes.

4.3 Revenue and expenditure attribution

For each sector-year with detailed components the balance is an identity, $B_{i,t} = R_{i,t} - E_{i,t}$, and so is its change:

$$\Delta B_{i,t} = \Delta R_{i,t} - \Delta E_{i,t}. \tag{4}$$

Two conventions keep Equation (4) unambiguous. Changes are computed only between adjacent years inside a continuous source block, so no change spans the 1995-to-2000 subsector gap. And where a term's role in the balance change is what matters, we report its *contribution*, carrying the sign with which it enters: the contribution of expenditure is $-\Delta E$, not ΔE. This matters because a rise in expenditure reduces the balance, so placing ΔE beside ΔB invites adding two quantities of opposite sign. Contribution columns sum to ΔB; the plain change columns do not.

The attribution is also hierarchical rather than parallel. Where an episode is attributed to a subsector, the revenue and expenditure split reported beside it is that *subsector's* own, not the aggregate's, so both halves of the table describe the same entity.

Equations (2) and (4) are computed on different source families — the canonical balance panel and the detailed account panel. They therefore measure the same annual change slightly differently, and we carry the residual between them as its own column rather than reconciling it away.

4.4 Decomposing the Social Security balance change

For the Social Security Funds the same identity is taken one level further. The revenue side splits into social contributions and everything else across the whole detailed panel; the expenditure side splits into social transfers and everything else for the modern period, where component detail exists. Writing contributions to the balance change,

$$\Delta B_t^{SSF} = \underbrace{\Delta C_t + \Delta R_t^{oth}}_{\text{revenue}} \underbrace{-\Delta T_t - \Delta E_t^{oth}}_{\text{expenditure}}, \tag{5}$$

where C is social contributions, T social transfers paid, and the *oth* terms the respective remainders. Equation (5) closes exactly by construction. It locates an annual movement in the accounts; it does not explain it.

4.5 Offset metric

Writing the combined non-Social-Security balance as $B_t^{nonSSF} = B_t^C + B_t^{RL}$, the offset ratio is defined only when the signs make the comparison meaningful, that is when $B_t^{nonSSF} < 0$ and $B_t^{SSF} > 0$:

$$O_t = \frac{B_t^{SSF}}{|B_t^{nonSSF}|}. \tag{6}$$

$O_t = 1$ means the positive Social Security balance is exactly the size of the negative non-Social-Security balance. The ratio compares two recorded numbers. It is left undefined, and missing, in every year where the sign condition fails, and it is not a counterfactual: see Section 2.

4.6 Persistence

Each annual balance is classified by sign. We report, per subsector, the frequency of each sign, the mean and median balance ratio, the longest unbroken run of each sign, and the empirical one-year sign transition frequencies.

Two conventions follow from the 1995 splice. Magnitudes are computed inside each regime, because a mean that averages two vintages with different levels describes neither. Runs are *not* recomputed per regime: a run is a property of the uninterrupted series, and truncating it at a window boundary would report the length of the window rather than the length of the run. Transition frequencies are empirical frequencies over the transitions actually observed, not a fitted Markov model: no standard errors and no stationarity test are claimed, and a state that never occurs has no estimated row.

4.7 Piecewise-constant mean shifts

Fewer than fifty annual observations split across two regimes do not support a flexible change-point model. The specification is deliberately restrictive: a piecewise-constant mean with at most two breaks per regime, a minimum segment length of five years, and exact dynamic-programming minimisation of within-segment squared error. This is the mean-shift case of the multiple-structural-change problem treated by Bai and Perron (1998), and selecting the number of breaks by a Schwarz criterion follows Yao (1988).

We describe the selection rule as a *BIC-style penalized criterion* rather than as "the BIC", because the change-point literature uses several penalties and the choice is not neutral. Ours charges for segment means and break locations alike:

$$\mathrm{BIC}(k) = n \log\left(\frac{\mathrm{RSS}(k)}{n}\right) + (2k+1)\log n,$$

where k is the number of breaks. Charging for the locations is the conservative choice here: it penalises an additional break twice and makes the model less willing to claim one, which is what we want from a short sample. Zero breaks is always admissible.

Detection runs separately inside 1977–1994 and 1995–2025, so the known splice cannot be selected as an economic break.

Because a single selected date from eighteen or thirty-one observations is not determined by the data, we report it with a BIC margin over the next-best break count and with a sensitivity grid over the two tuning parameters. The results are in Section 6 and Appendix A, and the dates are described as candidates throughout.

5 Results

5.1 Long-run composition

Figure 1 plots the four balance ratios over the full window. The visual impression is of two series that track each other closely and two that stay near zero: General Government and Central Government are nearly indistinguishable for long stretches, while Regional and Local Government and the Social Security Funds move in a narrow band.

That closeness is itself the first result, and it can be stated as a number rather than left to the eye — but it has to be stated inside each statistical regime, for the same reason the magnitudes are. A correlation computed across the 1995 splice runs partly through a vintage change, and would attribute to co-movement what is in part a change of source family. Within the historical regime the General Government and Central Government ratios correlate at 0.973, with a median absolute gap of 0.53 percentage points of GDP; within the modern regime, at 0.976 and 0.55 points. That the two regimes agree is the substantive point, and it is a result rather than an

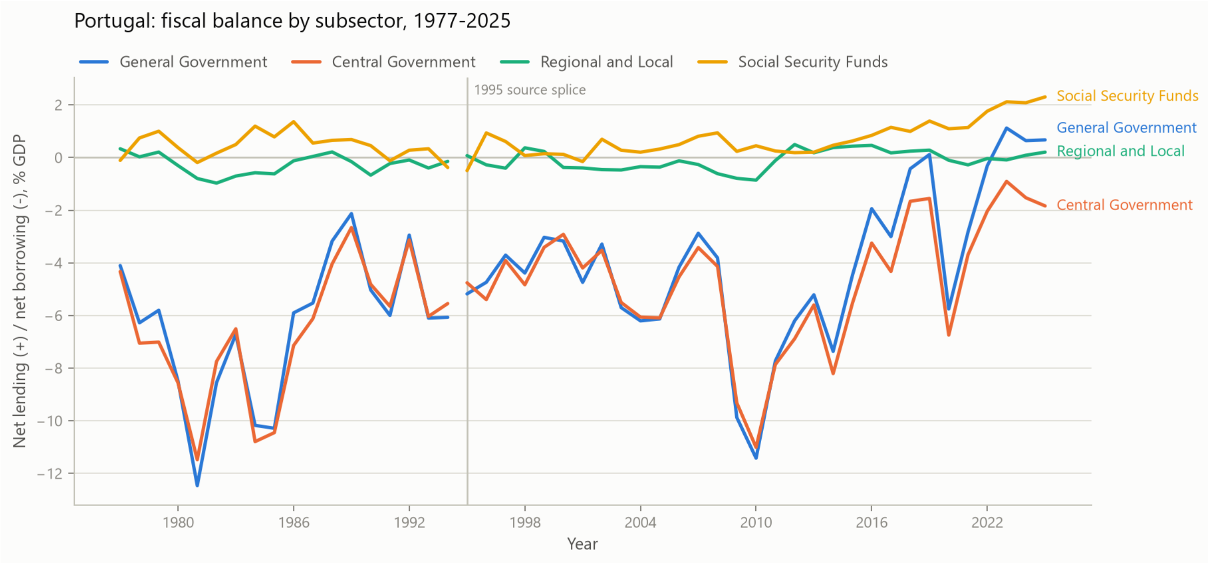

Figure 1: Net lending (+) and net borrowing (−) by subsector, 1977–2025, as a percentage of nominal GDP. The vertical rule marks the 1995 source splice; the series either side of it come from different source families and are not chained. Sources: Banco de Portugal / INE long series and INE via PORDATA.

assumption: the aggregate balance has been predominantly a Central Government quantity in level terms throughout, which is what makes the divergence at the right-hand edge of the figure notable rather than routine.

Figure 2 draws the identity directly, stacking the signed subsector contributions against the aggregate total. Where a tall stack sits under a shallow line, the subsectors moved in opposite directions and offset one another.

5.2 Persistence, and why magnitudes are regime-specific

Table 4 reports sign frequencies and magnitudes inside each statistical regime.

Table 4: Subsector composition of the general-government balance, by statistical regime. Magnitudes are reported inside each regime because the two differ enough that a pooled mean describes neither.

Regime	Subsector	N	Positive	Negative	Mean (% GDP)	Median (% GDP)
1977–1994 historical	General Government	18	0	18	-6.43	-6.03
1995–2025 modern	General Government	31	4	27	-4.04	-4.17
1977–1994 historical	Central Government	18	0	18	-6.61	-6.31
1995–2025 modern	Central Government	31	0	31	-4.67	-4.33
1977–1994 historical	Regional and Local	18	5	13	-0.27	-0.19
1995–2025 modern	Regional and Local	31	13	18	-0.09	-0.10
1977–1994 historical	Social Security Funds	18	14	4	0.46	0.47
1995–2025 modern	Social Security Funds	31	29	2	0.72	0.61

Three patterns are stable across both regimes. The Central Government balance is negative in every observation of the panel, all 49 of them. The Social Security balance is positive in 43 of 49, negative in 6, with a longest unbroken positive run of 24 years. Regional and Local Government is the only subsector that changes sign with any regularity, positive in 18 years, and it is small in both directions.

Magnitudes, by contrast, are strongly regime-dependent, and this is the reason the table is split. The aggregate balance averages -6.43% of GDP over 1977–1994 and -4.04% over 1995–2025. A pooled mean of -4.91% falls between the two and corresponds to no period in the data. Central

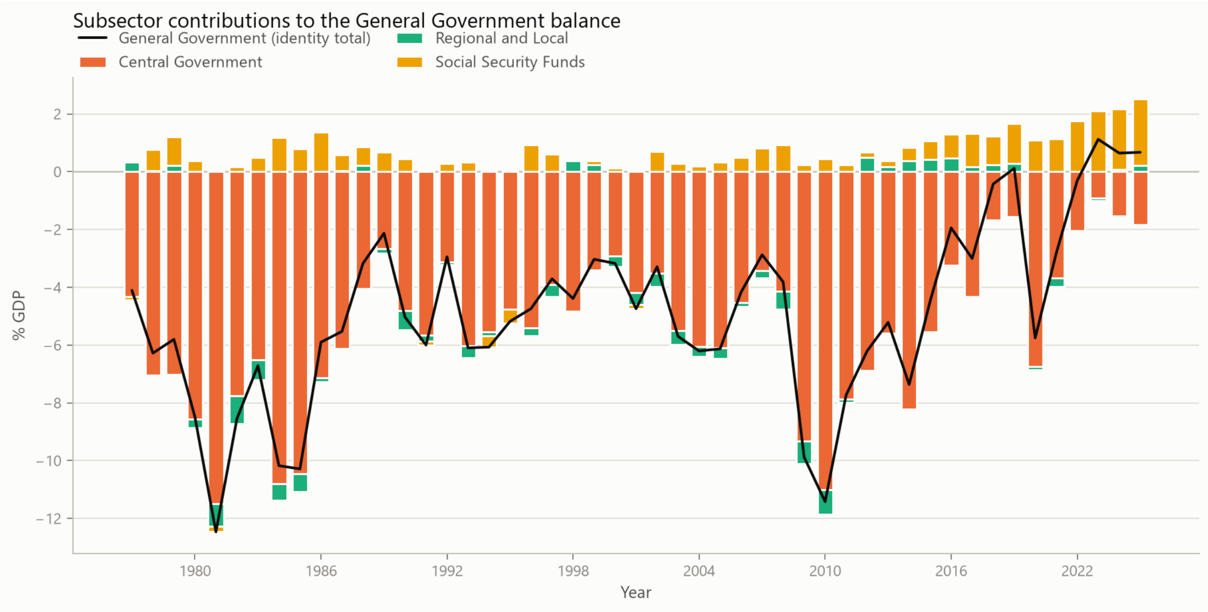

Figure 2: The identity of Equation (1) drawn: signed subsector contributions stacked against the General Government total, as a percentage of GDP. Positive and negative components are stacked separately from zero, and the General Government line gives their algebraic sum. The total visible span of a column is therefore the sum of the absolute contributions, not the aggregate balance; a tall span under a shallow line means the subsectors offset one another that year.

Government moves from -6.61% to -4.67% and the Social Security Funds from 0.46% to 0.72%.

Sign counts survive pooling far better than magnitudes. That is an empirical claim about this panel rather than a general property: a revision can in principle move a small balance across zero, and Section 3 checks whether it does here. Across both retained source overlaps the sign classifications are unchanged, which is why we pool the counts in the discussion above and never pool the means.

5.3 The years with a positive aggregate balance

The aggregate balance is positive in 4 of 49 years: 2019, 2023, 2024 and 2025. Table 5 gives the composition of each.

Table 5: Every year in which the aggregate balance is positive, with its subsector composition and the offset ratio.

Year	GG (M EUR)	Central (M EUR)	Regional/local (M EUR)	SSF (M EUR)	Non-SSF (M EUR)	Offset ratio
2019	249	-3,331	604	2,975	-2,727	1.091
2023	3,030	-2,447	-243	5,720	-2,690	2.126
2024	1,863	-4,424	256	6,031	-4,168	1.447
2025	2,059	-5,636	630	7,065	-5,006	1.411

The composition is the same in all 4, and it is worth being precise about which part of that is evidence. The empirical finding is that the combined non-Social-Security balance is *negative* in every one of them. Given that and $B_t^{GG} > 0$, the remaining two conditions — that $B_t^{SSF} > 0$ and that $B_t^{SSF} > |B_t^{nonSSF}|$ — follow from Equation (1) rather than constituting separate results. Reporting all three as findings would present a consequence of the identity three times.

What the identity does not fix is the magnitude, which is what the offset ratio of Equation (6) measures. In 2025 the identity reads

$$2,059 = -5,636 + 630 + 7,065 \quad \text{M EUR},$$

with a combined non-Social-Security balance of -5,006 million euro and an offset ratio of 1.411. In ratio terms the aggregate balance was 0.67% of GDP against a Social Security balance of 2.30%.

Figure 3 plots the ratio over the years in which it is defined.

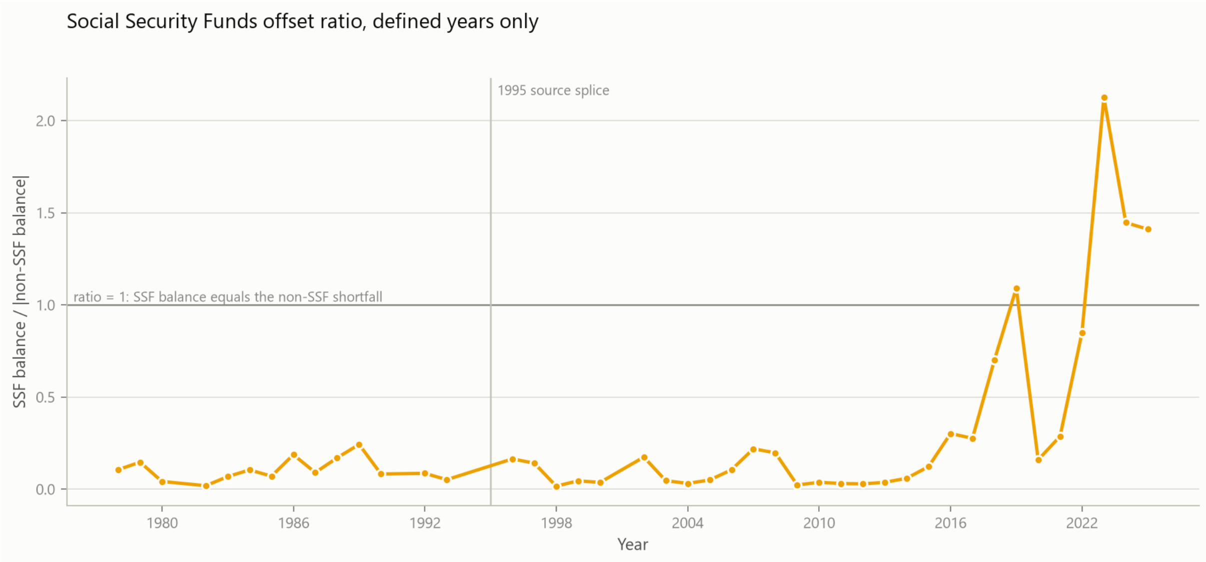

Figure 3: The offset ratio of Equation (6), for the years in which its sign condition holds. The reference line at one marks the level at which the Social Security balance equals the non-Social-Security shortfall. The ratio relates two recorded quantities and is not a counterfactual.

The ratio relates two published numbers, on the terms set out in Section 2.

5.4 Annual movements

Figures 4 and 5 attribute each annual change to the subsectors, on the GDP scaling of Equation (3). The two windows are drawn as separate panels and neither crosses 1995: a single panel spanning the splice would place a vintage revision among the economic movements and give it the same visual weight.

Table 6 ranks the 5 largest movements *inside each regime*. Ranking within a regime rather than across both is required by the same logic that splits the magnitudes in Table 4: each annual change is computed inside one source family and is sound, but ordering historical against modern episodes by size would compare two methodologies.

Table 6: The largest annual movements inside each statistical regime, ranked on the absolute change scaled by current-year GDP.

Regime	Rank	Year	Change (% GDP)	Central (M EUR)	Regional/local (M EUR)	SSF (M EUR)	Largest contributor
1977–1994 historical	1	1981	-5.40	-385	-48	-45	Central Government
1977–1994 historical	2	1984	-4.71	-922	-1	133	Central Government
1977–1994 historical	3	1980	-4.02	-233	-34	-29	Central Government
1977–1994 historical	4	1993	-3.28	-2,057	-210	46	Central Government
1977–1994 historical	5	1990	-3.26	-1,305	-271	-58	Central Government
1995–2025 modern	1	2009	-5.98	-8,939	-289	-1,268	Central Government
1995–2025 modern	2	2020	-5.88	-10,224	-811	-777	Central Government
1995–2025 modern	3	2011	3.92	5,937	1,342	-366	Central Government
1995–2025 modern	4	2015	2.63	4,283	121	321	Central Government
1995–2025 modern	5	2021	2.52	5,570	-394	270	Central Government

Ranking is within a regime rather than across both. Each annual change is computed inside one source family, so the changes are sound, but ordering historical against modern episodes by size would compare two methodologies. 1995 is excluded: that change straddles the splice in both panels.

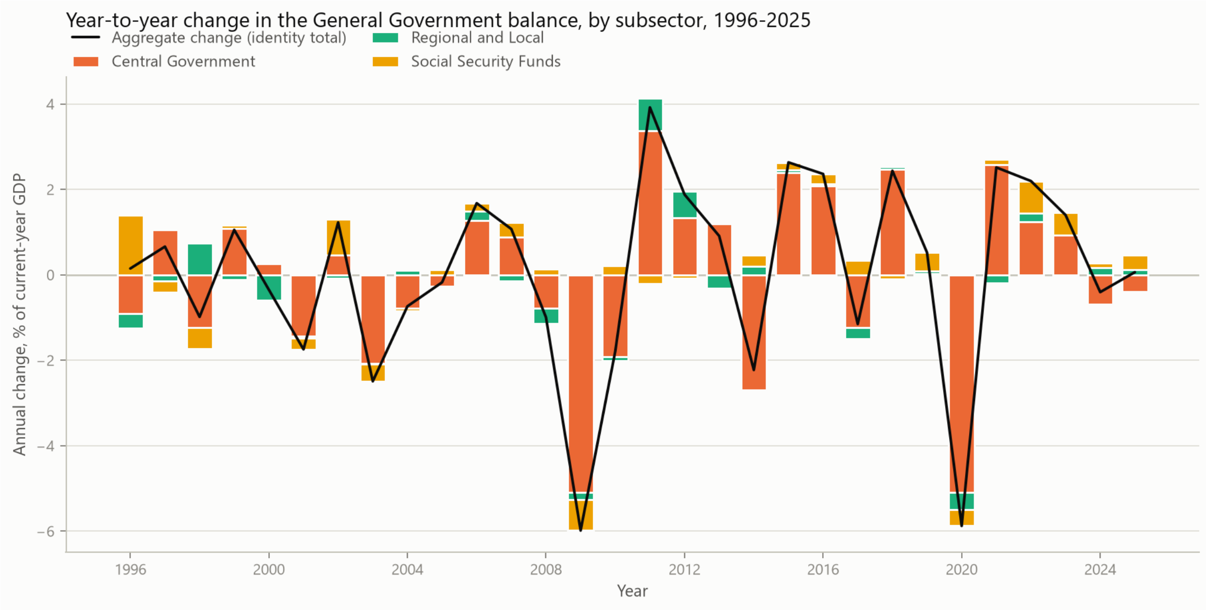

Figure 4: Annual change in the General Government balance attributed to subsectors, 1996–2025, as a percentage of current-year GDP. The black line is the identity total.

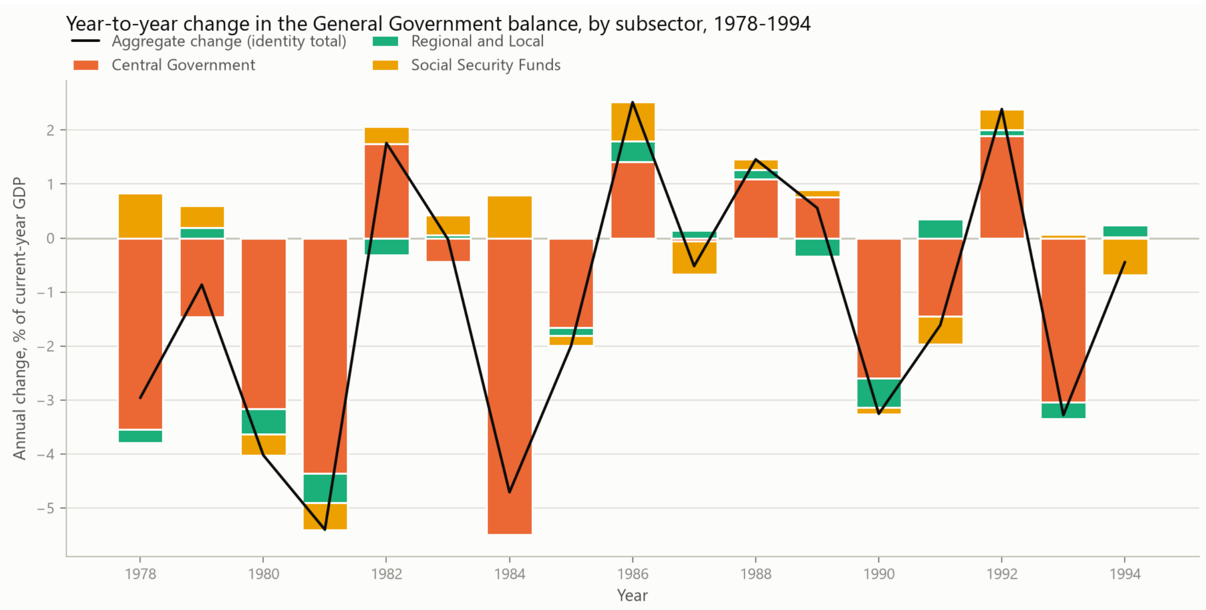

Figure 5: The same decomposition for 1978–1994, on the same scaling. The windows are plotted separately because no change is computed through the 1995 splice.

The scaling is what makes the exercise possible at all. On nominal euro the historical regime has no large episodes, because the amounts are small relative to modern GDP; on the GDP scale its movements are comparable to the modern ones.

The extremes are nevertheless reported by regime and never selected across both. In the modern regime the largest improvement is 2011 at 3.92 percentage points of GDP and the largest deterioration 2009 at -5.98; in the historical regime the largest deterioration is 1981 at -5.40. Naming a single "largest movement of the panel" would rank a historical episode against a modern one — comparable on the scaling, incomparable on the methodology — which is the same thing Table 4 refuses to do with the means.

The two regimes differ in a way the ranking exposes. In the historical regime the 5 largest movements are all deteriorations; the large improvements of that period are smaller in absolute size than its large deteriorations. The modern regime contains both, which is why it has a largest improvement to report and the historical regime does not.

Central Government accounts for the largest share of every episode in the table. Given that it also carries the level of the aggregate balance, at the within-regime correlations reported above, this is less surprising than it first appears.

Table 7 takes each episode one level further, splitting the dominant subsector's own movement into its revenue and expenditure changes. The split is of that subsector and not of the aggregate, which is what allows the two tables to be read together.

Table 7: The same episodes, with the subsector accounting for most of each move split into its own revenue and expenditure changes.

Regime	Year	Largest contributor	Its balance change (M EUR)	Δ revenue (M EUR)	Δ expenditure (M EUR)	Expenditure contribution (M EUR)
1977–1994 historical	1981	Central Government	-385	395	780	-780
1977–1994 historical	1984	Central Government	-922	616	1,538	-1,538
1977–1994 historical	1980	Central Government	-233	388	621	-621
1977–1994 historical	1993	Central Government	-2,057	-496	1,561	-1,561
1977–1994 historical	1990	Central Government	-1,305	1,866	3,172	-3,172
1995–2025 modern	2009	Central Government	-8,939	-3,467	5,473	-5,473
1995–2025 modern	2020	Central Government	-10,224	-3,748	6,475	-6,475
1995–2025 modern	2011	Central Government	5,937	1,209	-4,728	4,728
1995–2025 modern	2015	Central Government	4,283	1,580	-2,703	2,703
1995–2025 modern	2021	Central Government	5,570	6,855	1,285	-1,285

The split is of the named subsector, not of the aggregate. Expenditure enters the balance negatively, so the final column is minus the expenditure change; it is that column which adds to the revenue change to give the subsector's balance change.

Reading the signs matters here. Expenditure enters the balance negatively, so the final column reports minus the expenditure change; it is that column, added to the revenue change, which gives the subsector's balance change. The 2011 improvement, for instance, comes overwhelmingly from expenditure falling rather than from revenue rising, and the 2021 improvement from revenue rising while expenditure also rose.

5.5 Which accounts moved

Totals say whether an episode was driven by revenue or by expenditure. They cannot say which account moved, and two episodes with the same revenue–expenditure split can have entirely different contents. Table 8 takes the attribution one level further, to the components of the dominant subsector's own accounts.

The two source families resolve the accounts at different depths, and we do not force a common scheme on them. The modern workbooks separate four revenue and seven expenditure components; the historical series separates current from capital and no further. Each sector-year uses the finer scheme it reports, so the modern period is never coarsened to match the historical one, and the historical rows are visibly coarser rather than spuriously detailed.

The component view separates episodes that the totals make look alike. The modern regime's largest deterioration, 2009, is dominated by taxes at -4,248 million euro: a collapse on the

Table 8: The three largest component movements behind each episode, for the subsector that dominates it, ranked by the absolute size of the contribution.

Regime	Year	Side	Component	Change (M EUR)	Contribution (M EUR)
1977–1994 historical	1981	Expenditure	Current expenditure	702	-702
1977–1994 historical	1981	Revenue	Current revenue	397	397
1977–1994 historical	1981	Expenditure	Capital expenditure	78	-78
1977–1994 historical	1984	Expenditure	Current expenditure	1,529	-1,529
1977–1994 historical	1984	Revenue	Current revenue	591	591
1977–1994 historical	1984	Revenue	Capital revenue	25	25
1977–1994 historical	1980	Expenditure	Current expenditure	525	-525
1977–1994 historical	1980	Revenue	Current revenue	380	380
1977–1994 historical	1980	Expenditure	Capital expenditure	96	-96
1977–1994 historical	1993	Expenditure	Current expenditure	1,494	-1,494
1977–1994 historical	1993	Revenue	Current revenue	-486	-486
1977–1994 historical	1993	Expenditure	Capital expenditure	68	-68
1977–1994 historical	1990	Expenditure	Current expenditure	2,555	-2,555
1977–1994 historical	1990	Revenue	Current revenue	1,663	1,663
1977–1994 historical	1990	Expenditure	Capital expenditure	617	-617
1995–2025 modern	2009	Revenue	Taxes	-4,248	-4,248
1995–2025 modern	2009	Expenditure	Capital expenditure	1,571	-1,571
1995–2025 modern	2009	Expenditure	Social transfers	1,346	-1,346
1995–2025 modern	2020	Revenue	Taxes	-3,355	-3,355
1995–2025 modern	2020	Expenditure	Other current expenditure	3,122	-3,122
1995–2025 modern	2020	Expenditure	Capital expenditure	2,087	-2,087
1995–2025 modern	2011	Expenditure	Capital expenditure	-4,594	4,594
1995–2025 modern	2011	Expenditure	Interest	2,235	-2,235
1995–2025 modern	2011	Revenue	Taxes	1,984	1,984
1995–2025 modern	2015	Expenditure	Capital expenditure	-3,095	3,095
1995–2025 modern	2015	Revenue	Taxes	1,775	1,775
1995–2025 modern	2015	Expenditure	Social transfers	1,060	-1,060
1995–2025 modern	2021	Revenue	Taxes	3,485	3,485
1995–2025 modern	2021	Revenue	Sales and other current revenue	1,616	1,616
1995–2025 modern	2021	Revenue	Capital revenue	1,460	1,460

The two source families resolve the accounts at different depths, so each sector-year uses the finer scheme it reports rather than being coarsened to a common one: four revenue and seven expenditure components in the modern period, current against capital in the historical one. Expenditure contributions are negated, so a falling capital expenditure appears as a positive contribution.

revenue side rather than an expenditure surge. Its largest improvement, 2011, is dominated by capital expenditure at 4,594 million: an expenditure retrenchment, and specifically a capital one, appearing as a positive contribution because a fall in expenditure raises the balance. The 2021 improvement, by contrast, is led by revenue components throughout.

The historical episodes cannot be read at that resolution, and the table shows why: their rows carry only current and capital. That is a property of the source, and the alternative — assigning historical movements to modern component names — would be fabrication.

5.6 Social Security mechanisms

We now turn to the second research question. Two distinct layers have to be kept apart, and the value of the exercise lies in the separation.

The national-accounts layer. The ESA 2010 Social Security Funds sector is the entity that appears in Equation (1). In 2025 social contributions were 67.07% of its total revenue. Figure 6 plots that share over the panel.

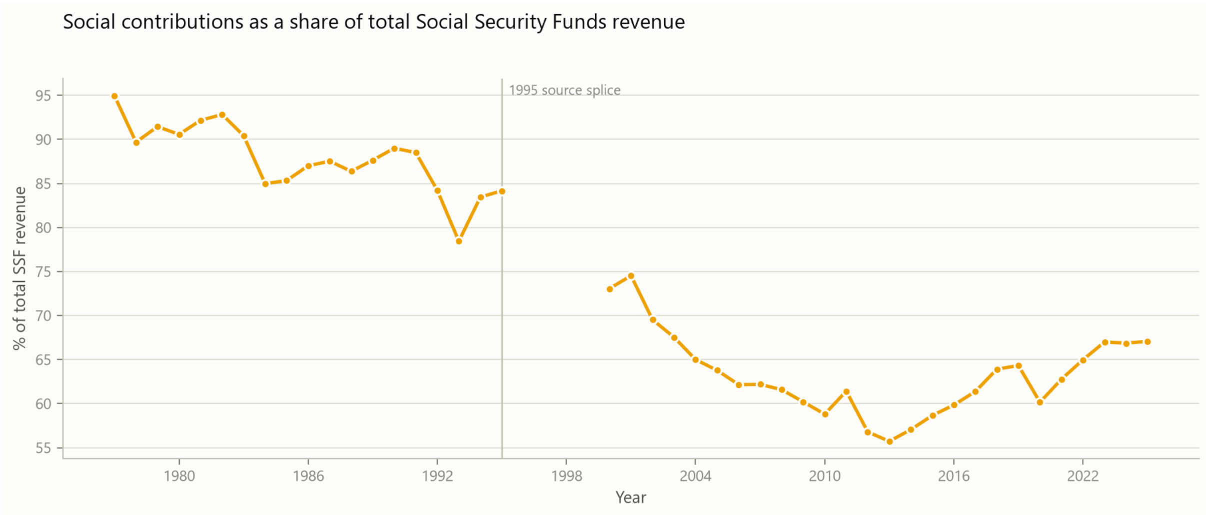

Figure 6: Social contributions as a share of total Social Security Funds revenue. The line breaks at 1996–1999, where subsector components do not exist in either source family.

Where the annual change comes from. The contribution share describes composition; it does not say what moved the balance. Equation (5) does, and Table 9 reports it for the recent years.

Table 9: Contributions to the annual change in the Social Security balance. Each column carries the sign with which the term enters the balance, so the four add to the change.

Year	Change in balance (M EUR)	Social contributions (M EUR)	Other revenue (M EUR)	Social transfers (M EUR)	Other expenditure (M EUR)
2018	-208	1,166	-368	-836	-170
2019	941	1,484	653	-1,145	-51
2020	-778	311	2,197	-1,449	-1,837
2021	270	1,430	-454	-794	88
2022	1,831	2,289	138	-1,525	929
2023	1,421	2,806	337	-1,653	-69
2024	310	2,557	1,339	-3,017	-569
2025	1,034	2,468	1,087	-2,003	-518

A rise in social transfers appears as a negative contribution, because expenditure reduces the balance. The plain expenditure change would have the opposite sign and could not be added to the revenue terms.

In 2025 the balance rose by 1,034 million euro. Social contributions contributed 2,468 million and other revenue 1,087 million; social transfers contributed -2,003 million and other expenditure

-518 million. The pattern across the recent years is the same in each: contributions are the largest positive term and social transfers the largest negative one, with the balance change being the residual of two large and partly offsetting movements. Figure 7 shows the full modern window.

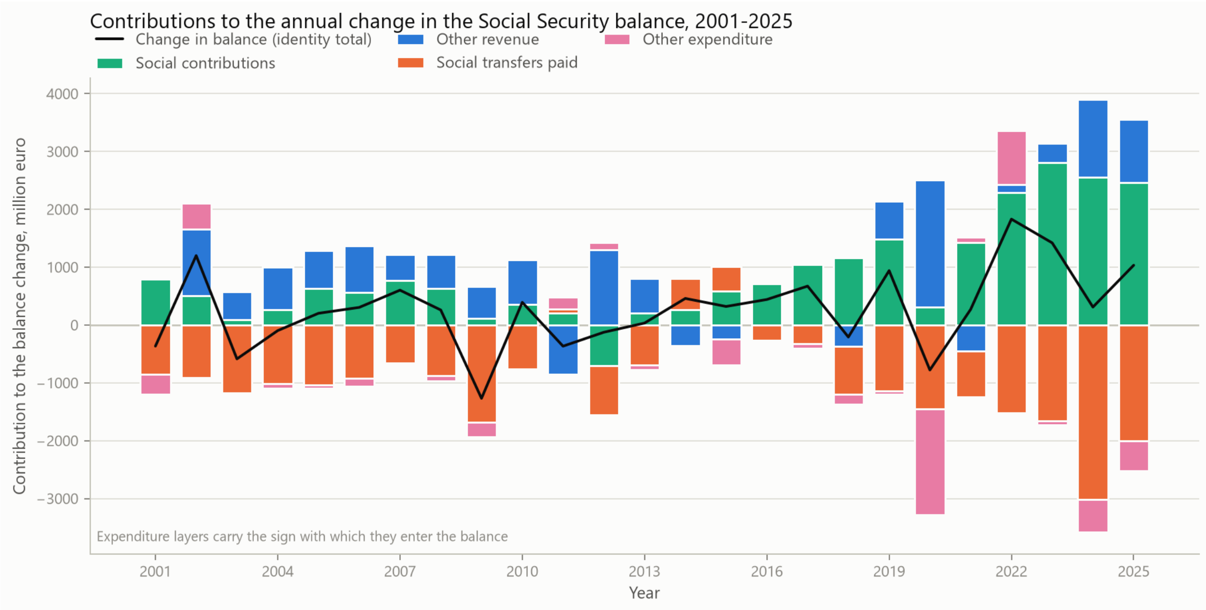

Figure 7: Contributions to the annual change in the Social Security balance, 2001–2025, in million euro. Each layer carries the sign with which it enters the balance, so layers above zero raised the balance that year and layers below zero reduced it; the black line is the identity total.

This is consistent with what the Council reports for 2025 from the budget accounts, namely that revenue growth exceeded expenditure growth with contributions the main driver (Portuguese Public Finance Council, 2026c). Our decomposition is on the national-accounts boundary rather than the budget one, so the two are not the same arithmetic; the agreement in direction and in the dominant term is nevertheless a useful external check.

What this decomposition does not do is relate the movement to anything outside the fiscal accounts. It says which account moved, not what the account moved with. Section 7 takes that step, relating contributions to the aggregate employee wage bill that is their principal observable reference base. Even there the account remains descriptive: a contributory balance responds to employment, wages, contribution rates, entitlement rules and demographics at once, and none of those is identified.

The budget-execution layer. The CFP publishes the system decomposed into the Previdential system, the Social Protection of Citizenship system and special regimes. These are budget-execution aggregates on a different accounting boundary. Figure 8 shows them.

The movement is concentrated in one system. The Previdential balance goes from 2,333 million euro in 2019 to 6,712 million in 2025, while the Citizenship balance stays small in both directions, standing at -55 million in 2025. Since contributions are the dominant revenue item of the Previdential system, and since they are also the dominant positive term in the national-accounts decomposition above, the two layers place the movement in the same part of the accounts.

Why the two layers must not be interchanged. This is a methodological result, and it is the one we would most want a reader to carry away. Table 10 places the two side by side.

In 2025 the ESA 2010 balance was 7,065 million euro while the three budget systems summed to 6,657 million, a difference of 408 million. Across the 7 overlapping years the difference ranges from 120 to 436 million euro. It is never zero.

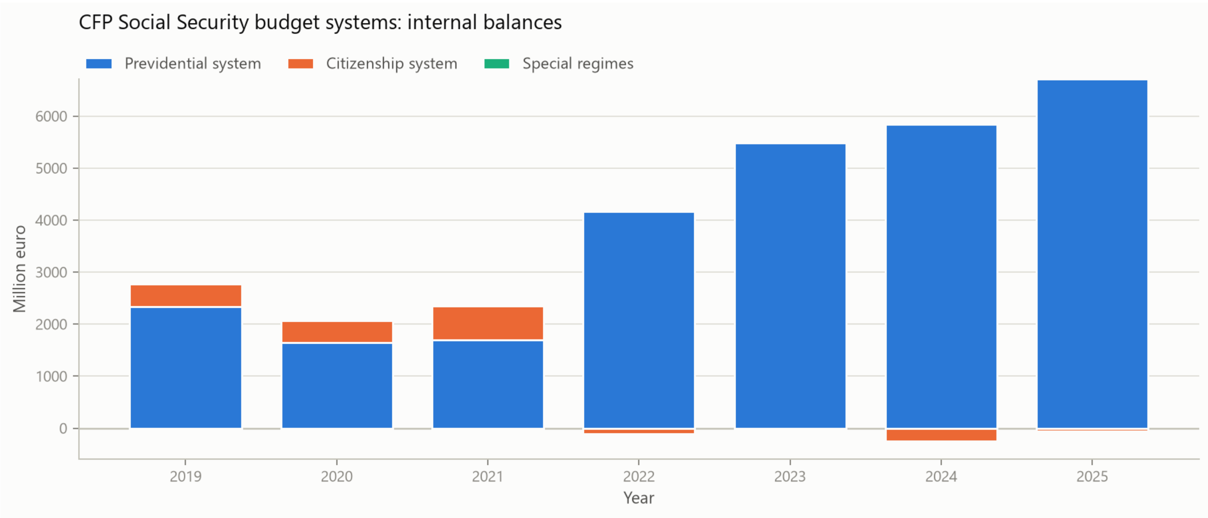

Figure 8: CFP internal Social Security balances by system, in million euro. The movement over the published window is concentrated in the Previdential system.

Table 10: Two accounting boundaries for Social Security, reported side by side. The columns are never added, netted or reconciled.

Year	ESA 2010 (M EUR)	Budget systems (M EUR)	Previdential (M EUR)	Citizenship (M EUR)	Difference (M EUR)
2019	2,975	2,776	2,333	443	199
2020	2,198	2,072	1,649	423	126
2021	2,468	2,348	1,695	653	120
2022	4,299	4,062	4,172	-110	237
2023	5,720	5,486	5,485	1	234
2024	6,031	5,595	5,840	-245	436
2025	7,065	6,657	6,712	-55	408

Figure 9 makes the scale problem visible: on a shared axis the two series are nearly identical, and the difference is only legible on an axis of its own.

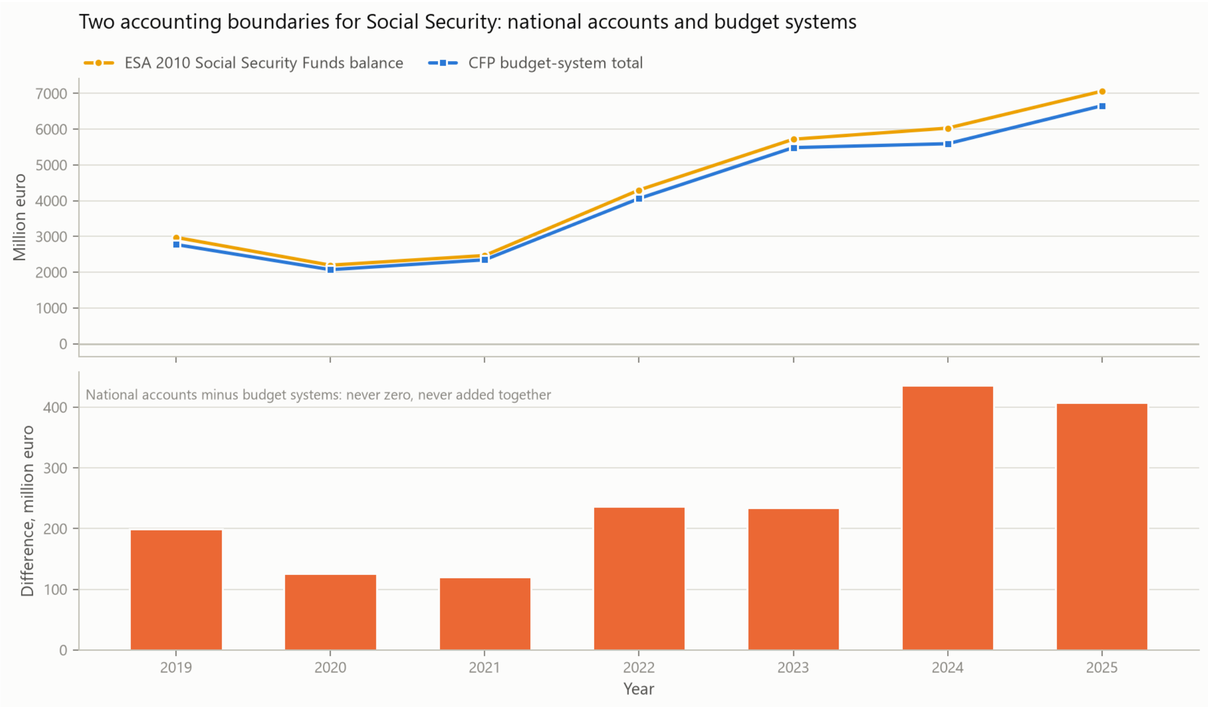

Figure 9: The two Social Security accounting boundaries. Upper panel: both balances on a shared axis, where they are almost indistinguishable. Lower panel: the difference on its own axis. The two are never added, netted or reconciled here.

The consequence is practical. A Social Security figure quoted from the budget documents is not the figure that enters the national-accounts identity, and substituting one for the other introduces an error of a few hundred million euro without any signal that it has happened. We therefore report both and add neither.

5.7 A persistently negative B.9 is not a persistently negative primary balance

The Central Government balance is negative in all 49 years of the panel. Read alone, that invites a strong conclusion: that the subsector is in deficit on every measure. The detailed accounts do not support it.

The primary balance is reconstructed as $PB_{i,t} = B_{i,t} + I_{i,t}$, where I is interest expenditure, and checked against the published primary-balance row. Table 11 contrasts the two sign frequencies.

Table 11: Headline against primary balance sign frequencies, over the sector-years for which interest expenditure is published.

Subsector	N	Headline < 0	Primary > 0	Mean interest (% GDP)	Peak interest (% GDP)
General Government	49	45	22	4.16	7.89
Central Government	45	45	15	4.17	7.73
Regional and Local	45	29	20	0.13	0.62
Social Security Funds	45	6	39	0.00	0.04

Over the 45 years for which interest expenditure is published, the Central Government headline balance is negative in 45 — all of them — while the primary balance is positive in 15: 1986, 1987, 1988, 1989, 1990, 1991, 1992, 1994, 1995, 2016, 2018, 2019, 2023, 2024 and 2025. Those years are not confined to the recent period; they run from 1986 to 2025.

The counts pool across the splice on the terms established above, and the split by regime is informative in its own right: the primary balance is positive in 9 of 19 historical years against 6 of 26 modern ones, so the positive years are if anything a historical phenomenon rather than a recent one. Interest is the arithmetic separating the two series, and it is a magnitude, so it is reported by regime and not pooled: it averaged 5.52% of GDP in the historical regime against 3.19% in the modern one. The pooled 4.17% and the 7.73% peak therefore describe the historical regime far better than the modern one, and we quote them only as panel-wide reference points. Figure 10 shows the primary balance crossing zero repeatedly while the headline balance never does.

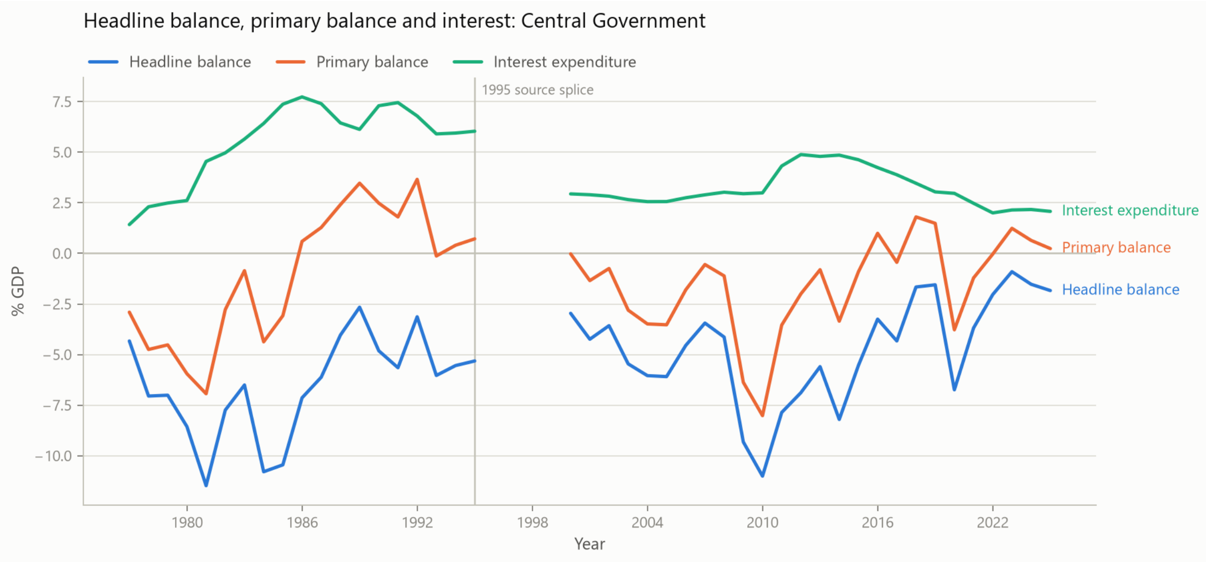

Figure 10: Central Government headline balance, primary balance and interest expenditure, as shares of GDP. The primary balance crosses zero repeatedly; the headline balance does not. The line breaks at 1996–1999.

Both statements are descriptive and they do not conflict. The correct summary is that the Central Government B.9 is negative throughout the panel *and* that its primary balance is positive in a non-trivial minority of the observed years. We call the second quantity the primary balance and not an underlying or structural balance: it removes interest and nothing else, and in particular it is neither cyclically adjusted nor net of temporary measures.

Symmetrically, a positive primary balance is not a sustainability result. It excludes interest by construction and therefore says nothing about the debt path, which depends on precisely the burden it removes.

6 Robustness and Accounting Boundaries

6.1 The 1995 splice

The splice is the single largest threat to the level results, and it is handled by restriction rather than by adjustment. Both 1995 vintages are retained; the revision between them is non-zero for every subsector; no level statistic is averaged across the boundary; and no annual change is computed through it. The regime split of Table 4 is the form in which magnitudes are reported precisely because a pooled magnitude would embed the splice silently.

Sign-based results are less exposed to the splice than magnitudes, and Section 3 checks that rather than asserting it: sign classifications are unchanged across both source overlaps this panel retains. The persistence results of Section 5 therefore carry more weight than the means beside them, and we have kept the two visually separate for that reason.

6.2 How firm are the change-point dates?

The piecewise-constant mean model of Section 4 selects break dates, and a selected date reads as more determined than it is — particularly here, where the penalised criterion follows Yao (1988) in form but is one of several the change-point literature admits (Bai and Perron, 1998). Two guards are reported.

The first is a BIC margin over the next-best break count, which says how decisively the selection beat its alternatives. The second is a sensitivity grid: detection re-run over 12 combinations of the two tuning parameters, neither of which is estimated from the data. A date that survives only one combination is a property of that choice rather than of the series.

Table 12 reports the grid summary, and Appendix A the full detail.

Table 12: Sensitivity of the detected mean shifts to the two tuning choices, over twelve specifications per series.

Regime	Subsector	Modal breaks	Modal dates	Share at modal dates	Distinct date sets
1977–1994 historical	General Government	1	1986	0.83	2
1995–2025 modern	General Government	2	2009, 2015	0.42	4
1977–1994 historical	Central Government	1	1987	1.00	1
1995–2025 modern	Central Government	2	2009, 2016	0.67	2
1977–1994 historical	Regional and Local	0	–	0.83	2
1995–2025 modern	Regional and Local	2	2012	0.33	6
1977–1994 historical	Social Security Funds	0	–	1.00	1
1995–2025 modern	Social Security Funds	1	2017	0.67	3

The grid is not a formality: it changes what may be claimed. In 6 of the 8 series the preferred specification returns the modal set of dates. In the remaining series it does not, and for those the selected dates follow the tuning choice. The least stable case is Regional and Local in the modern regime, where the modal dates hold in only 33% of the grid across 6 distinct date sets.

Accordingly we describe every date as a candidate rather than a finding, and no break year should be quoted from this paper without its share. We also attach no historical cause to any of them, and note that a five-year minimum segment means a shift in the last four years of the sample cannot be detected at all.

6.3 Intergovernmental transfers and the location of a balance

Because the subsectors are consolidated, a transfer between them changes the components and not the aggregate. This makes the *location* of a balance within general government partly a function of the transfer convention, which matters for interpreting composition results.

The historical source tables identify current and capital transfers between public administrations, which supports a purely mechanical sensitivity for 1977–1995:

$$B_{i,t}^{sens} = B_{i,t} - \left(T_{i,t}^{received} - T_{i,t}^{paid}\right).$$

We compute it and deliberately decline to label it an underlying or true balance, and we never use it to restate B.9. The reason is substantive rather than cautious: a transfer normally finances an expenditure responsibility assigned to the recipient, so removing the transfer while leaving the responsibility in place does not describe a coherent alternative arrangement. Which transfer discharges which statutory obligation is a legal question, not an accounting one. What the sensitivity quantifies is how much the recorded location of a balance depends on the convention — and it is reported for that purpose only.

The same reasoning is why no Social Security balance in this paper has State transfers netted out of it.

6.4 What the balance does not determine

Two limits on the reach of these results are worth stating explicitly, because both concern inferences a reader may reasonably be tempted to draw.

The balance does not determine the debt path. The change in debt equals minus the balance plus a stock-flow adjustment that absorbs financial-asset transactions, valuation effects and timing differences, and in several years in these data the adjustment is the larger of the two terms. Whether such residuals reflect accounting responses to fiscal rules (von Hagen and Wolff, 2006) or are smaller and less systematic than that literature suggested (Seiferling, 2013) is a live question on which we take no position. Its relevance here is narrower: a composition result about balances is not a result about debt.

The balance does not identify a mechanism. The decompositions of Section 4 locate a movement in the accounts; they do not explain it. Nothing here separates cyclical from discretionary movements, and no cyclical adjustment is attempted. A specification relating balances to nominal GDP growth exists in the repository and is reported there in an appendix with its weaknesses set out; it is deliberately not part of this paper's evidence, because its dependent variable and its regressor share a construction through nominal GDP and the fits are very low.

For the Social Security side specifically, that gap is closed rather than left open. Equation (5) shows that contributions are the dominant positive term in the recent balance changes, and the Council attributes the contribution growth to rising employment and average earnings (Portuguese Public Finance Council, 2026c). Section 7 models the contribution base directly, relating the change in contributions to the change in the aggregate wage bill $W_t = N_t\bar{w}_t$ rather than to aggregate nominal growth. The regressor there is a quantity the levy actually falls on, and the fit is an order of magnitude tighter than the specification criticised here.

7 Contributions and the Aggregate Employee Wage Bill

Every Social Security result to this point is internal to the fiscal accounts. Section 5 establishes that contributions are the largest positive term in the recent balance changes, and stops there: an accounting location, not an economic mechanism. This section supplies the link to something outside the accounts.

7.1 A reference base, and the ratio to it

Contributions are levied predominantly on wages, so the aggregate employee wage bill of the economy is the natural quantity to read them against: $W_t = N_t\bar{w}_t$, with N employees and $\bar{w}$ the average wage per employee.

It is a *reference base* and not the integral base of the levy, and the distinction carries through everything that follows. The contributions recorded in the national-accounts Social Security Funds sector are not all charged on employee wages. Some fall on other bases, the self-employed most obviously; some are imputed rather than levied at all. So W does not exhaust what C is charged on, and C/W is not the rate at which any contributor pays. What W does supply is the largest observable share of what contribution revenue is charged on, measured on the same statistical system as the fiscal accounts and available annually — which is what a decomposition needs and what no statutory rate schedule provides.

Writing the contributions-to-wage-bill ratio as $\tau_t = C_t/W_t$, the change in contributions decomposes exactly:

$$\Delta C_t = \underbrace{\tau_{t-1}\Delta W_t}_{\text{wage-bill component}} + \underbrace{W_{t-1}\Delta\tau_t}_{\text{ratio component}} + \underbrace{\Delta W_t\Delta\tau_t}_{\text{interaction}}. \tag{7}$$

The wage-bill component splits again by the same identity, into an employment component and an average-wage component. Both decompositions are exact. The interaction term is carried rather than dropped or shared between the factors, because either would make the decomposition inexact while looking tidier.

We call τ a ratio and never a rate, deliberately. "Rate effect" would invite the reading that a movement in τ reflects a change in the statutory levy, and the next paragraph explains why that reading would be wrong.

The wage bill and the employee count are the Portuguese national accounts compiled by Statistics Portugal, taken from the harmonised dissemination of those accounts (Statistics Portugal (INE), 2026b,a). We use wages and salaries (D.11) rather than compensation of employees (D.1), because D.1 already includes employers' social contributions and would place part of the numerator inside the denominator.

What τ is not. It follows from the preceding paragraph that τ is not a statutory contribution rate. Its numerator contains revenue raised on bases its denominator does not measure, so it moves with coverage, compliance and composition as well as with legislated rates. It stands at 0.265 in 2025 against a panel mean of 0.221, both well below the headline levy, which is itself a sign that the two are not the same object. A movement in τ is therefore a residual to be named, not a policy change to be inferred.

7.2 What moved contributions

Table 13: The annual change in Social Security contributions, ΔC, split into a wage-bill component, a ratio component and their interaction.

Year	ΔC	Wage-bill component	Ratio component	Interaction
2018	1,166	947	206	12
2019	1,484	1,014	443	26
2020	311	28	283	0
2021	1,430	1,268	152	10
2022	2,289	2,110	163	17
2023	2,806	2,685	107	13
2024	2,557	2,056	463	38
2025	2,468	1,870	560	38

The three components sum to the change. The interaction is carried rather than dropped or shared between the factors, because either would make the decomposition inexact while looking tidier.

In 2025 contributions rose by 2,468 million euro, which decomposes as

$$2,468 = \underbrace{1,870}_{\text{wage bill}} + \underbrace{560}_{\text{ratio}} + \underbrace{38}_{\text{interaction}} .$$

The reference base did most of the work, but not all of it.

A second and separate application of the same identity splits the movement in the wage bill itself. In 2025 the wage bill rose by 7,196 million:

$$7,196 = \underbrace{2,253}_{\text{employees}} + \underbrace{4,842}_{\text{average wage}} + \underbrace{102}_{\text{interaction}} .$$

The two decompositions must not be read as one nested inside the other, and the tables are separate for that reason. They have different totals — 2,468 million against 7,196 million — and the employment and average-wage terms are components of ΔW, not sub-components of the 1,870 million wage-bill term of ΔC. What connects them is the factor τ_{t-1}: the wage-bill term of the first identity is τ_{t-1} times the total of the second. The correct reading of the pair is that

Table 14: The annual change in the wage bill itself, ΔW, split into an employment component, an average-wage component and their interaction.

Year	ΔW	Employment component	Average-wage component	Interaction
2018	3,998	1,423	2,521	54
2019	4,225	478	3,722	25
2020	116	-1,400	1,544	-29
2021	5,071	852	4,172	47
2022	8,372	2,985	5,194	193
2023	10,572	2,000	8,383	189
2024	8,058	992	6,996	70
2025	7,196	2,253	4,842	102

This decomposes ΔW, not the wage-bill component of ΔC in Table 13. They are different quantities, and the columns here do not sum to that component.

the movement was predominantly in the reference base rather than in the ratio, and that within that base it came more from wages per employee than from employee numbers.

The interaction terms are small here, but they are stated rather than absorbed: a paper that insists the decomposition is exact should not present a two-term version of a three-term identity.

Figure 11 shows both decompositions over the modern period, and the two crises are where the mechanism is clearest. In the austerity years the wage bill contracts on both margins at once, employment and wages falling together. In 2020 the two margins move in opposite directions — employment falls while average wages rise — so the wage bill is almost flat and contributions rise anyway, on the ratio rather than the base. Recent years are the reverse of austerity: the base grows strongly, and average wages contribute more than employment.

7.3 A companion regression

Section 6 sets out why the nominal-GDP specification carries no weight, and one of its faults was that the regressor was an aggregate that merely correlates with the balance rather than a quantity the levy is charged on. That fault is repairable, and repairing it is the point of this specification.

Regressing the annual change in contributions on the annual change in the wage bill over 25 adjacent-year pairs gives a slope of 0.248 with a HAC standard error of 0.022 and an R^2 of 0.926. Three things are worth noting about that.

The slope is close to the average contributions-to-wage-bill ratio of 0.221. That is consistent with a comparatively stable ratio component over the sample, but it is not implied by the identity: Equation (7) would deliver $\beta = \bar{\tau}$ only if $\Delta\tau$ and the interaction term were uncorrelated with ΔW, which nothing here establishes. The agreement is therefore a descriptive fact about this sample rather than an arithmetic necessity.

The fit is an order of magnitude tighter than the nominal-GDP regressions, whose R^2 values span 0.07 to 0.18. That difference is the substantive argument for the specification, not a claim about its sophistication.

And no changes are computed across the 1995-to-2000 gap in subsector accounts. That matters more than it sounds: bridging the gap treats a five-year movement as annual, and doing so drives the slope to 0.13 and the R^2 to 0.40. A single contaminated observation was enough to destroy the relationship.

7.4 A symmetric alternative

Equation (7) evaluates each factor at its base-year level and carries the interaction explicitly. An equally standard convention evaluates each factor at the midpoint of the two years, which splits

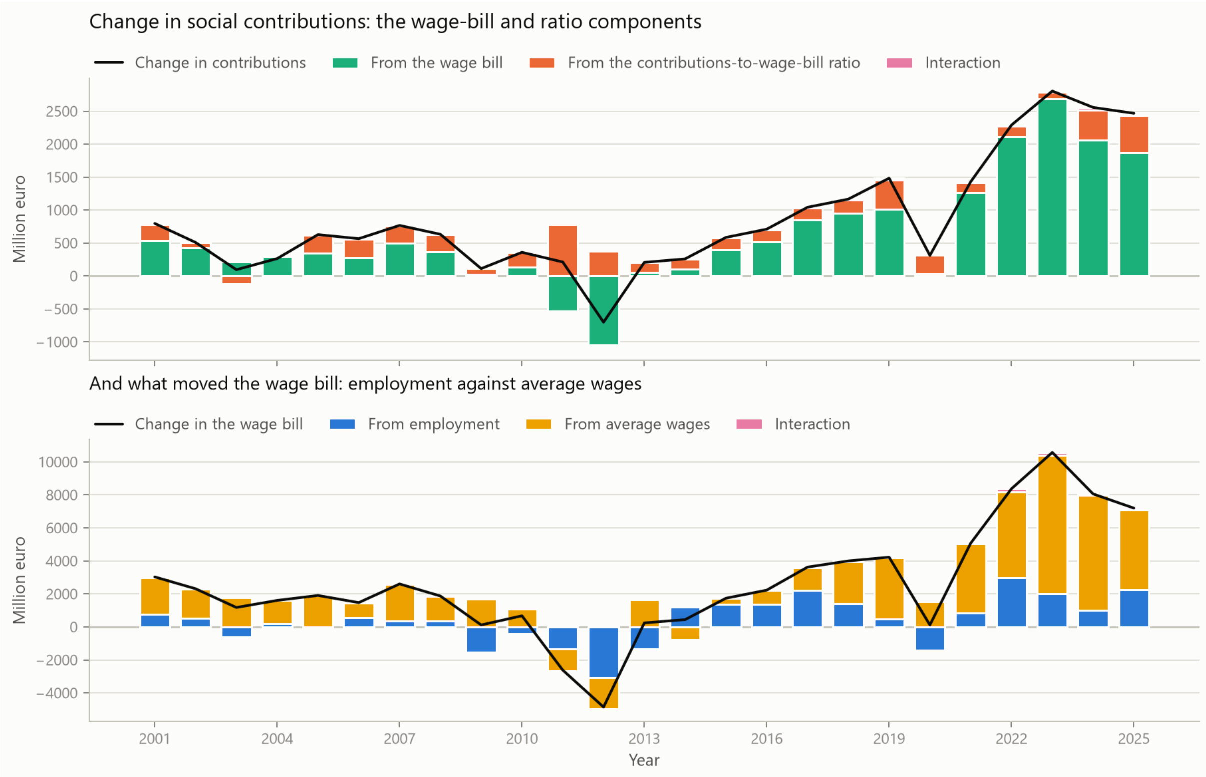

Figure 11: Contributions and their base, 2001–2025, in million euro. Upper panel: the change in contributions split into a wage-bill component, a ratio component and their interaction. Lower panel: the change in the wage bill split into an employment component, an average-wage component and their interaction. The two panels decompose different quantities and have different totals; both are exact, and the black line is the identity total in each.

the interaction evenly between them,

$$\Delta C_t = \tfrac{1}{2}\left(\tau_t + \tau_{t-1}\right)\Delta W_t + \tfrac{1}{2}\left(W_t + W_{t-1}\right)\Delta \tau_t,$$

and is exact in two terms rather than three. Neither convention is more correct. They answer slightly different questions, and which one is used is a choice worth checking rather than asserting.

Table 15: The same change under the symmetric convention, which evaluates each factor at the midpoint of the two years and thereby splits the interaction evenly between them.

Year	ΔC	Wage-bill component	Ratio component	Wage-bill share
2018	1,166	953	213	0.82
2019	1,484	1,027	456	0.69
2020	311	29	283	0.09
2021	1,430	1,273	157	0.89
2022	2,289	2,118	171	0.93
2023	2,806	2,692	114	0.96
2024	2,557	2,075	482	0.81
2025	2,468	1,889	579	0.77

Exact in two terms rather than three. Neither convention is more correct; the body reports the three-term form because carrying the interaction rather than distributing it is the more conservative presentation.

Under the symmetric convention the 2025 change splits into 1,889 million from the wage bill and 579 million from the ratio, against 1,870 and 560 under Equation (7). The difference between the two pairs is the interaction term, which the symmetric form distributes and the exact form reports; because the two differ by exactly half the interaction in each factor, the qualitative reading cannot turn on the choice. Over 2022–2025 the wage-bill share of the change runs from 77% to 96% on the symmetric convention. Earlier years in the table are not uniform — 2020 is the clear exception, when a flat wage bill left the ratio carrying almost the whole change — and that is a property of those years rather than of the convention. We report the three-term form in the body because carrying the interaction rather than distributing it is the more conservative presentation, and because a reader can always distribute an interaction that is reported and cannot recover one that is not.

7.5 What this does and does not establish

The decomposition of Equation (7) is exact and descriptive. It partitions a recorded movement; it does not show that the wage bill produced it, and it says nothing about why employment or wages moved. The regression quantifies a co-movement between two first differences and claims no identification.

What it does establish is that the recent growth in Social Security contributions is overwhelmingly associated with movement in the reference base rather than in the ratio to it, and that the base's own movement has lately come more from wages per employee than from the number of employees. Because the wage bill is a reference base and not the integral base of the levy, that is a statement about the largest observable component of what contributions are charged on, not about the whole of it. That is a mechanism in the accounting sense: it names the quantity whose movement the contributions followed. It is not a causal account of why that quantity moved.

8 A European Benchmark: Is This Composition Unusual?

Everything to this point describes one country. It establishes how Portugal's general-government balance is composed and how persistent that composition is; it cannot establish whether the composition is unusual, because a single-country study has no comparison set. That distinction matters for how the results should be read: a subsector pattern that is common across Europe invites a different explanation from one that is peculiar to Portugal.

ESA 2010 requires the same subsector breakdown from every reporter, so the comparison is available (Eurostat, 2025; European Central Bank, 2025). We take net lending and net borrowing by subsector for 1995–2025 from Eurostat and repeat the domestic definitions on it.

8.1 Construction

Three choices matter, and each is a place where a careless benchmark would mislead.

The non-Social-Security aggregate must include state government. Portugal has no S.1312 tier, so its aggregate is central plus local. Germany, Spain, Austria, Belgium and Switzerland do have one, and omitting it would leave their identity open and understate their non-Social-Security deficits. We therefore define $B^{nonSSF}_{c,t} = B^{S.1311}_{c,t} + B^{S.1312}_{c,t} + B^{S.1313}_{c,t}$, treating a missing state tier as contributing nothing because the tier does not exist rather than because a value is unknown. With that definition the identity closes to published rounding for every reporter.

Ratios are computed in national currency. Eurostat publishes shares of GDP to one decimal, which is an unusable denominator for a ratio: a non-Social-Security balance printed as -0.2 could lie anywhere in a band wide enough to move the offset ratio by a quarter of its value. The levels carry far more precision, and the ratio is currency-invariant. We additionally require the denominator to be at least 0.5% of GDP in absolute size before reporting a ratio, so that a near-zero denominator cannot manufacture a large one.

Coverage is made comparable. A country-year enters only if the four required sectors — the aggregate, central government, local government and social security funds — are all reported. The state tier is required conditionally rather than absolutely: a country that reports S.1312 anywhere in its record must report it in that year too, while a country that never reports it, Portugal among them, is not excluded for lacking a tier it does not have. Requiring the tier unconditionally would drop every unitary reporter; not requiring it at all would admit federal country-years whose non-Social-Security aggregate is incomplete, and it is exactly those years in which the identity would fail to close. A reporter then enters the summary only with at least fifteen complete years, because a frequency computed over a handful of years is not comparable with one computed over thirty. This leaves 28 reporters over 1995–2025, with Portugal contributing 31 years.

Two remarks on what the comparison is worth. Eurostat's Portuguese rows agree with our domestic panel to rounding, which is an independent check on the extraction of Section 3 as well as the basis for the comparison. And this is a distribution, not a test: locating Portugal within a spread of accounting compositions says how common that composition is, not why any country's takes the form it does.

8.2 Where Portugal sits

Table 16 reports the composition for every reporter and Table 17 locates Portugal in each distribution.

The answer is not uniform across the paper's findings, and that is the substantive result of this section.

The Central Government deficit is not unusual. Portugal records a Central Government deficit in 100% of its reported years, which places it at the 68th percentile — but 9 of the 28 reporters also record one in every single year. A persistently deficit-running central tier is a common European pattern, not a Portuguese peculiarity. Read on its own, the domestic finding that Central Government is negative in all 49 years would invite the opposite impression.

Table 16: Subsector composition across reporters, ordered by the frequency of a Social Security surplus. Sign frequencies are shares of the years each reporter covers completely.

Reporter	N	Central < 0	SSF > 0	Mean SSF (% GDP)	Surplus years	of which non-SSF < 0	Median offset
FI	31	0.774	1.000	2.45	11	6	0.540
LU	31	0.645	1.000	1.65	24	10	0.892
DK	31	0.355	0.935	0.05	21	0	0.008
CY	31	0.806	0.935	2.23	9	4	0.446
PT	31	1.000	0.935	0.71	4	4	0.141
IT	31	1.000	0.839	0.18	0	0	0.064
EL	31	0.903	0.839	0.97	6	3	0.206
EE	31	0.613	0.839	0.25	12	2	0.306
IS	28	0.536	0.821	0.16	10	0	0.043
SE	31	0.581	0.806	0.19	14	2	0.225
SK	31	1.000	0.742	0.42	0	0	0.061
BG	31	0.613	0.710	0.17	13	2	0.095
RO	31	1.000	0.710	0.13	0	0	0.084
CH	30	0.533	0.700	0.16	17	5	0.058
LV	31	0.968	0.677	0.35	2	0	0.316
LT	31	1.000	0.613	0.25	4	3	0.419
DE	31	0.774	0.613	0.04	8	2	0.085
NL	31	0.806	0.581	0.17	8	4	0.310
IE	31	0.516	0.581	0.09	16	3	0.167
BE	31	1.000	0.581	0.06	3	2	0.072
AT	31	0.968	0.548	0.02	2	0	0.056
HU	31	1.000	0.548	0.02	0	0	0.045
SI	31	0.935	0.516	-0.05	3	1	0.043
PL	31	1.000	0.484	0.24	0	0	0.263
HR	31	0.871	0.452	-0.12	4	1	0.124
FR	31	1.000	0.419	-0.20	0	0	0.072
CZ	31	0.935	0.387	-0.04	4	0	0.051
ES	31	0.903	0.355	-0.27	3	0	0.464

The non-Social-Security aggregate is central plus state plus local government, so federal reporters are treated consistently with unitary ones. Reporters with fewer than fifteen complete years are excluded, because a frequency over a handful of years is not comparable with one over thirty.

Table 17: Portugal's position in each cross-country distribution. The percentile is the share of reporters below Portugal's value.

Metric	Portugal	Cross-country median	Minimum	Maximum	Percentile
Share of years with a Central Government deficit	1.000	0.903	0.355	1.000	68
Share of years with a Social Security surplus	0.935	0.689	0.355	1.000	82
Share of years with an aggregate surplus	0.129	0.129	0.000	0.774	39
Mean Social Security balance (% GDP)	0.706	0.168	-0.271	2.448	82
Median offset ratio, where defined	0.141	0.110	0.008	0.892	54

The Social Security surplus is unusual, in both frequency and size. Portugal records a Social Security surplus in 93.5% of years against a cross-country median of 68.9%, at the 82th percentile. Its mean Social Security balance is 0.71% of GDP against a median of 0.17%, again at the 82th percentile. On both measures Portugal sits in the upper tail.

The typical offset magnitude is ordinary. Portugal's median offset ratio, over the years in which it is defined, is 0.141 against a cross-country median of 0.110, at the 54th percentile. In a typical year the Portuguese Social Security surplus offsets no more of the non-Social-Security deficit than elsewhere. What distinguishes Portugal is the recent years, in which the ratio exceeds one, not the central tendency.

That conclusion rests on the denominator floor, which is a choice rather than a property of the data, so it is worth checking whether the conclusion survives other choices. Sweeping the floor from 0.25 to 1.00 percent of GDP moves Portugal's percentile only between 50 and 57: the position stays near the middle of the distribution throughout. Had the percentile moved sharply with the floor, the finding would have been an artefact of where we set it.

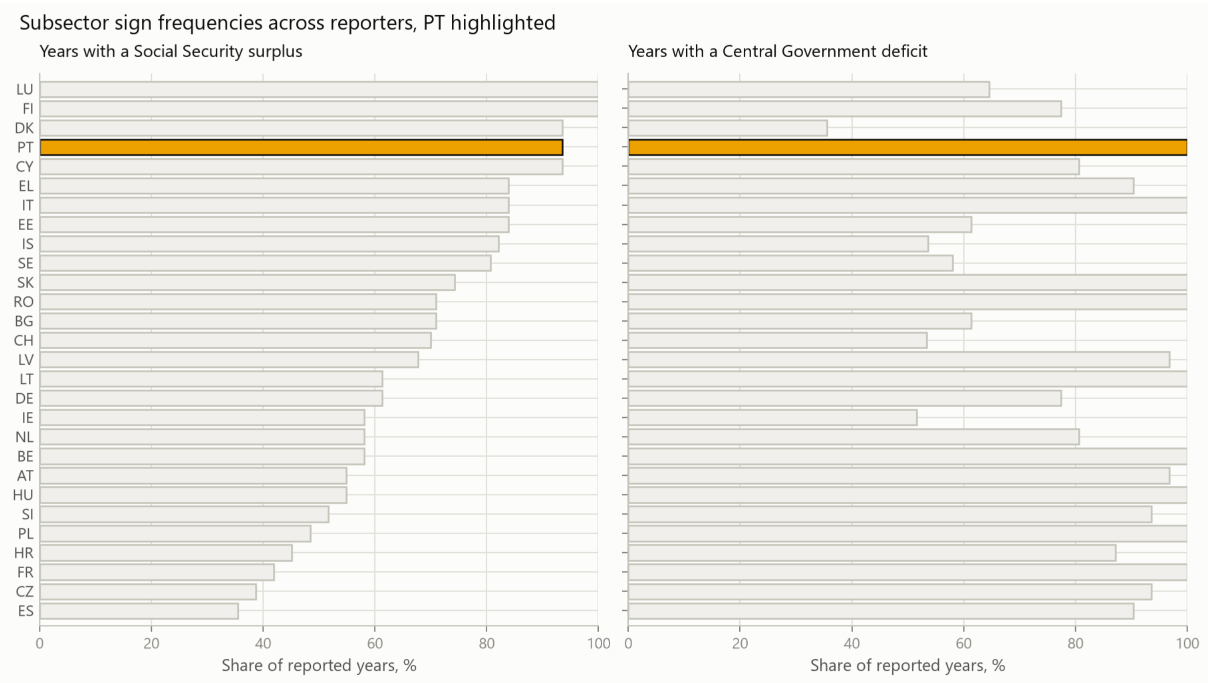

Figure 12: Subsector sign frequencies across reporters, Portugal highlighted, ordered by the frequency of a Social Security surplus. Left: share of reported years with a Social Security surplus. Right: share with a Central Government deficit. Portugal is in the upper tail of the first and shares the maximum of the second with several reporters.

8.3 The composition of a surplus year

The sharpest comparison is not a sign frequency but the structure of the surplus years themselves, because that is the paper's headline composition restated as a cross-country question: when a country records an aggregate surplus, is its non-Social-Security balance in deficit?

Across the 22 reporters with at least one aggregate surplus, that composition holds in 54 of 198 surplus country-years, or 27%. It is therefore a minority pattern: most European surpluses are recorded with the non-Social-Security tiers also in surplus, or close to balance.

Portugal is the exception in the strict sense. Of the 22 reporters, 1 record a negative non-Social-Security balance in *every* one of their aggregate-surplus years, and Portugal is among them, at 4 of 4.

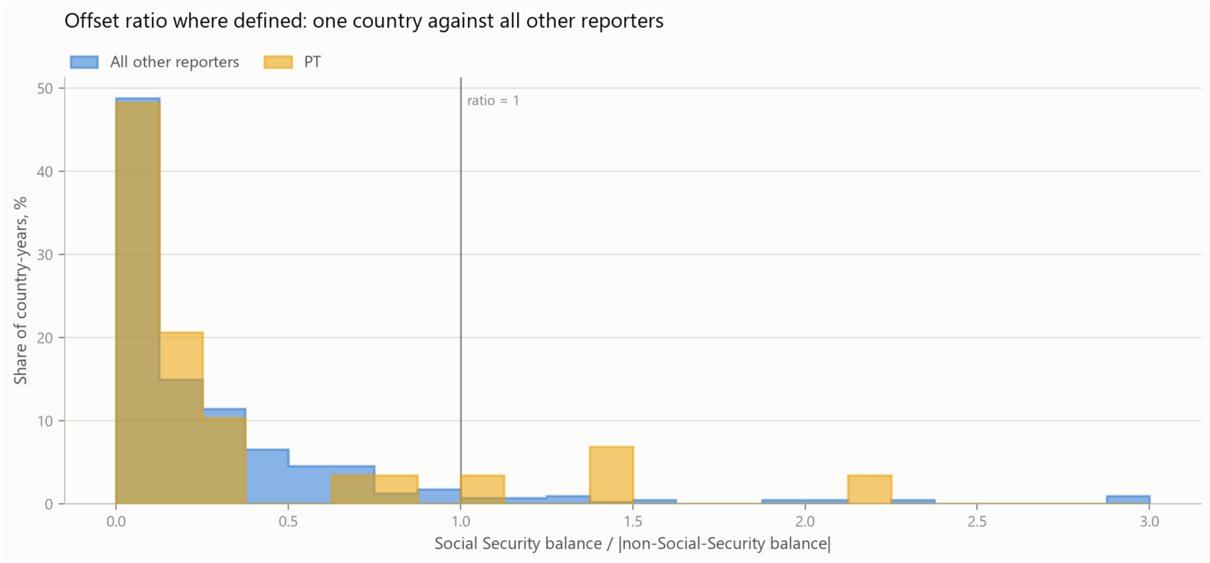

Figure 13: Offset ratio where defined: Portugal against all other reporters pooled, as a share of each group's defined country-years. Values above three are clipped to the last bin. The Portuguese distribution is not shifted relative to the others; its recent years sit in the right tail of both.

The pooled figure weights by country-year, so a reporter with two dozen surplus years counts for more than one with three. Weighting by country instead asks a different and equally reasonable question — of a country's own surplus years, what share pair with a negative non-Social-Security balance? — and the two agree: the country-weighted median is 25%, with quartiles at 4% and 49%, against the pooled 27%. Portugal sits at the 95th percentile of that distribution. Because the two weightings can disagree and neither is obviously correct, both are reported.

That result needs its sample stated plainly alongside it. Portugal has only 4 surplus years, so "all of them" is four observations, and several reporters with more surplus years have high but not unanimous shares. The defensible statement is therefore the conjunction: the composition is a minority pattern in Europe, Portugal exhibits it in every surplus year it has recorded, and the number of such years is small. We would not read a stronger claim than that from 4 observations.

8.4 What the benchmark changes

It changes the reading of two results and confirms a third.

The Central Government finding is weaker than it looks in isolation: a permanently deficit-running central tier is normal in Europe. The Social Security finding is stronger: both the frequency and the size of the Portuguese surplus sit in the upper tail. And the composition of the surplus years is genuinely distinctive, subject to the sample caveat above.

None of this identifies a cause. Countries differ in how they assign contributory schemes between tiers, in whether they operate a state-government level, in the maturity of their pension systems and in how they route transfers between subsectors — and the benchmark holds none of that constant. It establishes that the Portuguese composition is unusual in a specific, stated respect, which is what the rest of the paper could not do.

9 Conclusion

Portugal's general-government balance, decomposed across institutional subsectors over 49 years, has a composition that is asymmetric and remarkably stable. The Central Government balance is negative in every one of the 49 observations. The Social Security balance is positive in 43, with a longest unbroken run of 24 years. Regional and Local Government is the only subsector that changes sign with any regularity, and it is small in both directions. The aggregate tracks Central Government closely inside each statistical regime, correlating at 0.973 before the 1995 splice and 0.976 after it. The aggregate is positive in 4 years, and in each of them the combined non-Social-Security balance is negative — from which, given the identity, the rest of that year's composition follows.

Against 28 European reporters, the composition is distinctive in one specific respect and unremarkable in another. A permanently deficit-running central tier is common, so the Central Government finding is weaker than it appears in isolation. The Social Security surplus sits in the upper tail on both frequency and size. And Portugal is the only reporter whose every aggregate-surplus year combines that surplus with a negative non-Social-Security balance — a composition present in 27% of European surplus years, though Portugal's own count is only 4.

On the Social Security side, most of the recent increase in contributions is algebraically associated with growth in the aggregate employee wage bill rather than with movement in the contributions-to-wage-bill ratio. Of the 2,468 million euro rise in 2025,

$$2,468 = 1,870 + 560 + 38,$$

the three terms being the wage-bill component, the ratio component and their interaction. The wage bill itself rose by 7,196 million over the same year, and that is a second and separate identity:

$$7,196 = 2,253 + 4,842 + 102,$$

in employees, average wage and interaction. The two share a factor but not a total, and the terms of the second are components of ΔW rather than parts of the 1,870 million wage-bill component of the first. Read together they say that the movement was predominantly in the reference base rather than in the ratio, and that within the base it came more from wages per employee than from employee numbers. That is a mechanism in the accounting sense — it names the quantity the contributions followed — and not a causal account of why that quantity moved.

Two results cut against readings that the aggregate series invites.

The first concerns Central Government. Its balance is negative throughout the panel, but its primary balance is positive in 15 of the 45 years for which detailed accounts exist — 9 of 19 before the 1995 splice and 6 of 26 after it, so the positive years are if anything a historical phenomenon rather than a recent one. A persistently negative B.9 does not imply a persistently negative primary balance, and interest is the arithmetic that separates them: it averaged 5.52% of GDP in the historical regime against 3.19% in the modern one.

The second concerns Social Security. The ESA 2010 Social Security Funds sector and the CFP budget-execution systems are different accounting objects whose balances differ in every one of the 7 years they overlap, by between 120 and 436 million euro. Quoting one where the other belongs introduces an error of that order with no signal that it has occurred.

9.1 What this paper does not establish

The decomposition is an accounting identity. It reallocates a recorded quantity and estimates nothing, and three inferences do not follow from it.

It does not license a counterfactual. That a positive Social Security balance exceeds a negative non-Social-Security balance in a given year says those two published numbers stand in a particular ratio. It says nothing about what the aggregate would have been under a different institutional arrangement, because that counterfactual would move contribution rates, entitlements and transfers together and the identity models none of them.

It does not attribute responsibility. A subsector accounting arithmetically for most of a movement has not been shown to have produced it.

It does not speak to sustainability. Neither the headline nor the primary balance determines the debt path, which also depends on stock-flow adjustments that are the larger term in several years of these data.

It does not explain why Portugal's composition differs. The benchmark establishes that it does, in a stated respect, while holding no institutional feature constant.

Two further limits are properties of the data rather than of the method. Magnitudes are regime-specific and cannot be pooled across the 1995 splice; the pooled aggregate mean of -4.91% of GDP describes neither the -6.43% of 1977–1994 nor the -4.04% of 1995–2025. And 2024 and 2025 are provisional at source: they are the years discussed in most detail here and the years most likely to be restated by a later vintage.

9.2 Extensions

Two extensions follow naturally.

The contribution base of Section 7 stops at the aggregate wage bill. Splitting it by sector, contract type or earnings band would say which part of the base carried the growth, and the same identity would apply unchanged; the data required are published but are not in the bundle assembled here.

Extending the benchmark of Section 8 would sharpen it. It currently compares compositions while holding nothing else constant, so it cannot separate a Portuguese peculiarity from the consequence of an institutional choice made elsewhere: whether a country operates a state-government tier, how it assigns contributory schemes between tiers, how mature its pension system is, and how it routes transfers between subsectors. Conditioning on those would turn a distribution into a comparison.

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A Change-Point Sensitivity in Full

Section 6 summarises the sensitivity of the piecewise-constant mean shifts. This appendix records the specification and the full grid, so the summary can be checked rather than taken.

A.1 Specification

For a balance ratio $y_1, \ldots, y_n$ inside one statistical regime, the model partitions the index range into $k + 1$ contiguous segments and fits a constant to each. The partition minimises total within-segment squared error,

$$\min_{0=b_0<b_1<\cdots<b_k<b_{k+1}=n} \sum_{j=0}^{k} \sum_{t=b_j+1}^{b_{j+1}} \left(y_t - \bar{y}_{[b_j,b_{j+1}]}\right)^2,$$

solved exactly by dynamic programming rather than by a greedy or binary-segmentation heuristic. Selection over k uses

$$\mathrm{BIC}(k) = n \log\left(\frac{\mathrm{RSS}(k)}{n}\right) + (2k+1)\log n,$$

where the parameter count $2k+1$ charges for $k+1$ segment means and k break locations. Charging for the locations is the conservative choice: it penalises an additional break twice and makes the model less willing to claim one. Zero breaks is always admissible.

Two tuning parameters are not estimated: the minimum segment length and the maximum number of breaks. The baseline sets them to five years and two breaks per regime.

A.2 The grid

Detection is re-run over the product of

$$\text{minimum segment} \in \{4,5,6,7\} \quad \text{and} \quad \text{maximum breaks} \in \{1,2,3\},$$

giving 12 specifications for each of the 8 regime-subsector series. Three quantities are persisted for each series: the modal number of breaks and the share of the grid selecting it, the modal set of dates and the share of the grid returning exactly that set, and the number of distinct date sets the grid produced.

The last of these is the most informative single number. A series whose grid returns one date set is telling us something about the series; a series whose grid returns six is telling us about the tuning parameters.

A.3 Reading the result

6 of 8 series have a preferred specification that agrees with the modal grid outcome. Where the two disagree, the modal column is the more reliable summary and the preferred date should not be quoted alone. The least stable series is Regional and Local in the modern regime, at 33% modal agreement across 6 distinct date sets.

Two further limits apply regardless of the grid. A minimum segment length of five years means a shift within the last four years of the sample is undetectable by construction. And selecting a mean shift does not imply the series is piecewise-constant; it is the best fit inside a restricted model class, chosen because the sample is too short to support a richer one.

The full per-specification grid, the BIC ladder over admissible break counts and the stability summary are persisted as `outputs/tables/structural_break_sensitivity.csv`, `outputs/tables/structural_break_bic_ladder.csv` and `outputs/tables/structural_break_stability.csv`.

B Data and Code Availability

B.1 What is archived

The accompanying repository retains the raw source workbooks as downloaded, a SHA-256 digest of each, the source-specific intermediate extractions, the canonical processed panels, every calculated analysis table, the figures, and the executed notebooks with their outputs stored in place. No stage of the chain from a published workbook to a number in this paper is missing.

B.2 How this paper relates to the technical report

The repository contains two documents and they have different jobs.

The technical report is generated end to end: its prose is a thin restatement of the persisted artefacts, it carries every analysis the pipeline computes, and no part of it is authored by hand. It is the place to look for material this paper omits — the fixed-capital-formation diagnostic, the debt and stock-flow reconciliation, the full transition matrices, the intergovernmental-transfer sensitivity and the descriptive macroeconomic co-movement.

This paper is authored. Its argument, its structure and its prose are written, and it carries only the evidence that argument needs.

What the two share is the arithmetic. Every table in this paper is written by the pipeline from a persisted artefact, and every number in its prose is a LaTeX macro generated from one. A sentence here cannot hold a value the data no longer support: if an artefact changes, the macro changes with it, and a macro that ceases to exist fails the build rather than rendering silently as nothing. The authored text lives in `paper/sections/`; the generated inputs live in `paper/generated/` and are never edited by hand.

B.3 Reproduction

```
poetry install
make all          # pipeline, notebooks, tests
make paper        # this document
```

The pipeline is deterministic: the same bundled inputs produce the same outputs, and the outputs are committed, so a reader can check a number without running anything.

B.4 Known limitations of the archive

The bundled sources are a single vintage. 2024 and 2025 are provisional at source and a later vintage will restate them; reproducing this paper against newer files will therefore reproduce the method but not necessarily every figure. The vintage of each retained file is recorded in Table 1 for that reason.

The 4 missing years of subsector detailed accounts, 1996–1999, are absent from both source families. They are not imputed here and cannot be recovered from the bundled data.